

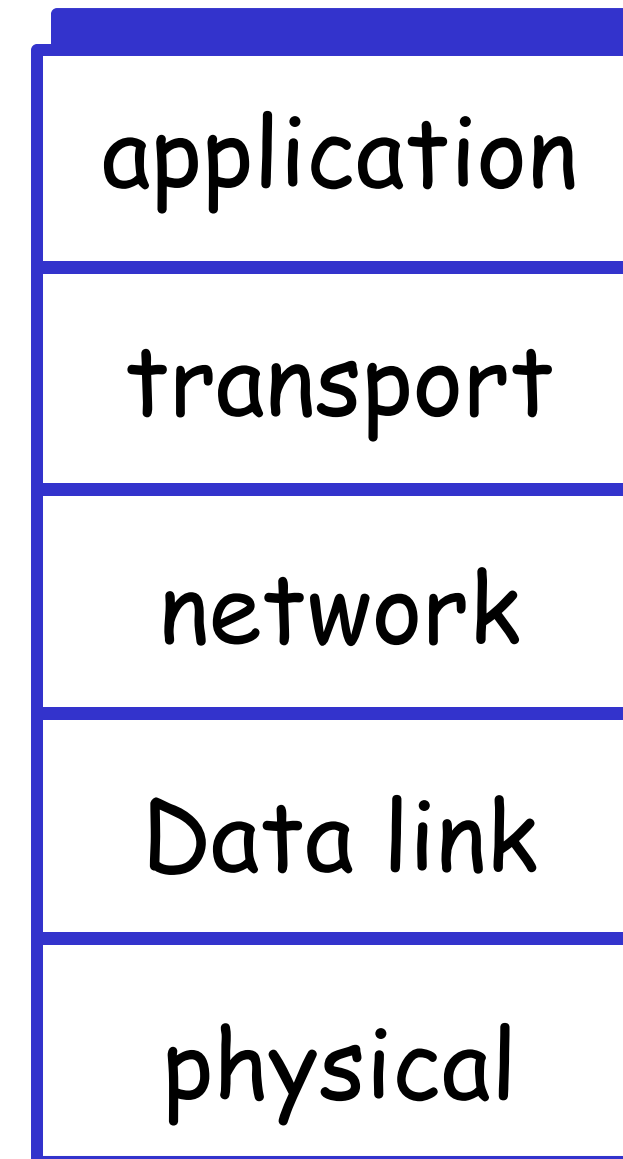
Data Link Layer Recap



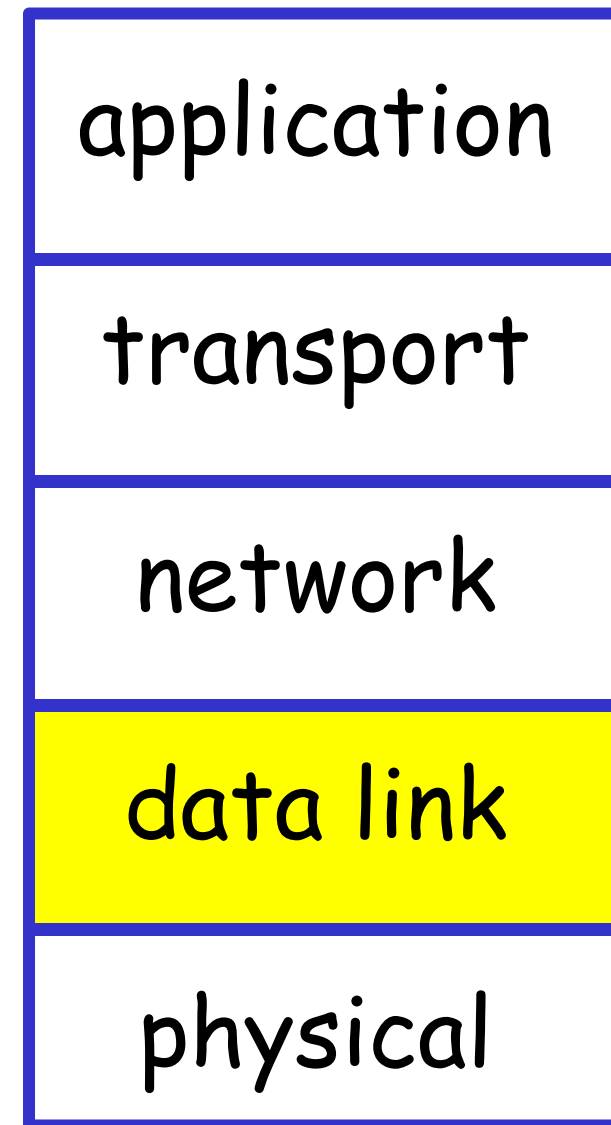
Lecture 5.0
Supporting Document [Optional]

Internet Protocol Stack

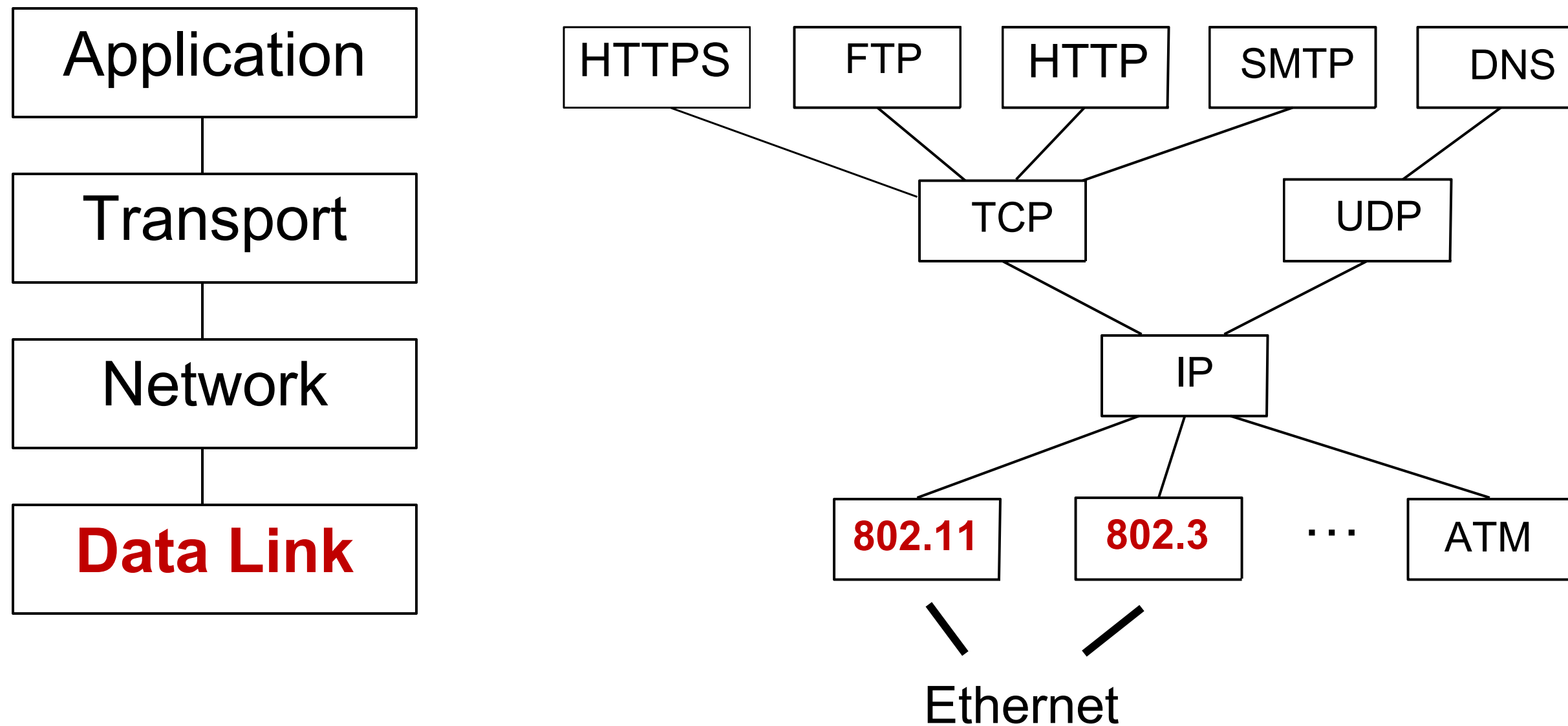
- **Application:** supporting network applications
 - FTP, SMTP, HTTP
- **Transport:** data transfer between processes
 - TCP, UDP
- **Network:** routing of datagrams from source to destination
 - IP, routing protocols
- **Data Link:** data transfer between neighboring network elements
 - Ethernet, WiFi
- **Physical:** bits "on the wire"
 - Coaxial cable, optical fibers, radios



Data Link Layer



Data Link layer



The main function of the link layer is **link-local delivery**: getting packets from one side of the link to the other.

Two kinds of data link layers

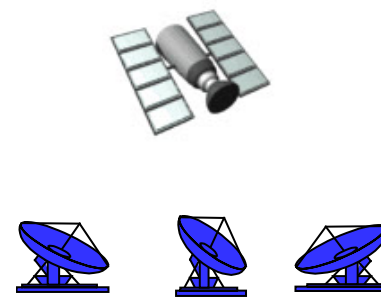
- **point-to-point**
 - Switched Ethernet link between switch and host
 - PPP for dial-up Internet access
- **broadcast (shared medium)**
 - Shared Ethernet
 - 802.11 wireless LAN



shared wire (e.g.,
cabled Ethernet)



shared RF
(e.g., 802.11 WiFi)



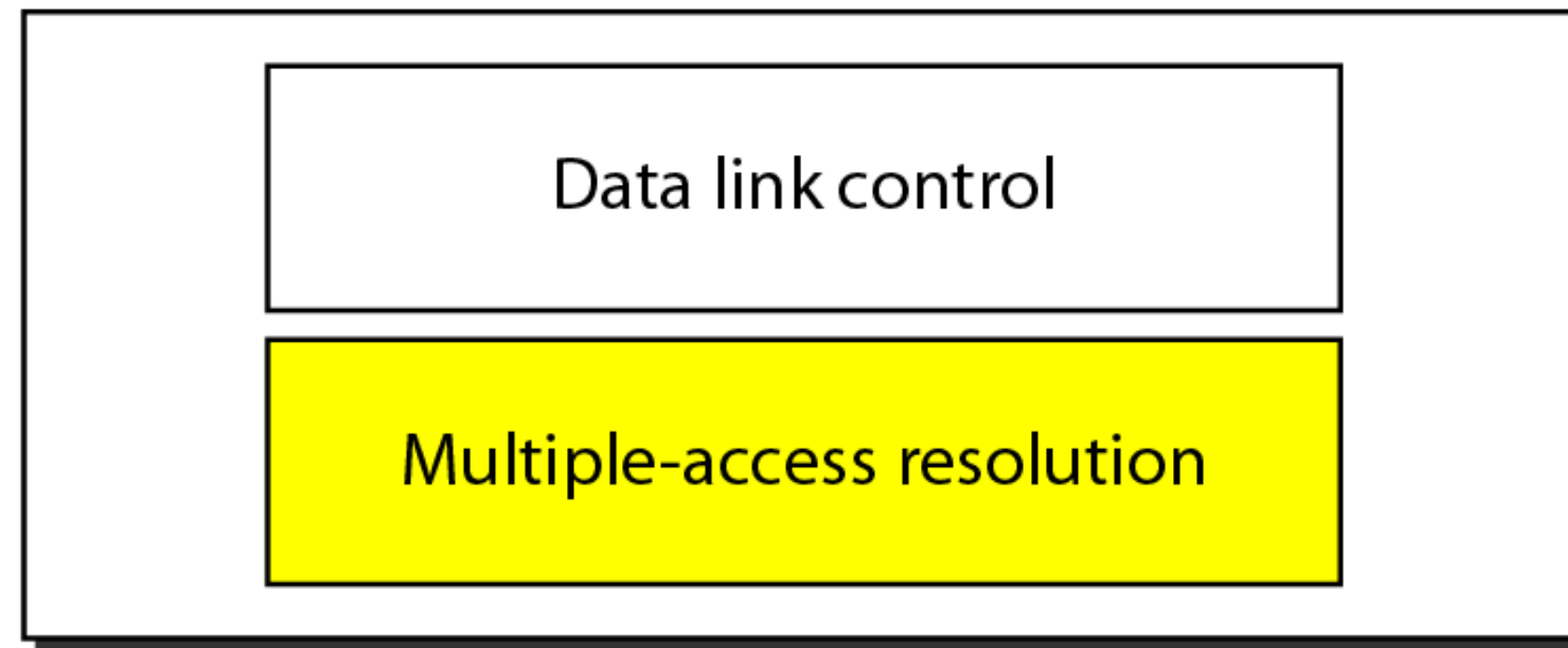
shared RF
(satellite)



humans at a
cocktail party
(shared air, acoustical)

Figure 1.1 *Data link layer divided into two functionality-oriented sublayers*

Data link layer



Multiple Access protocols

Multiple Access protocols

- single shared broadcast channel
- two or more simultaneous transmissions by nodes: interference
 - **collision** if node receives two or more signals at the same time

multiple access protocol

- distributed algorithm that determines how nodes share channel, i.e., determine when node can transmit

Ideal Multiple Access Protocol

Broadcast channel of rate R bps

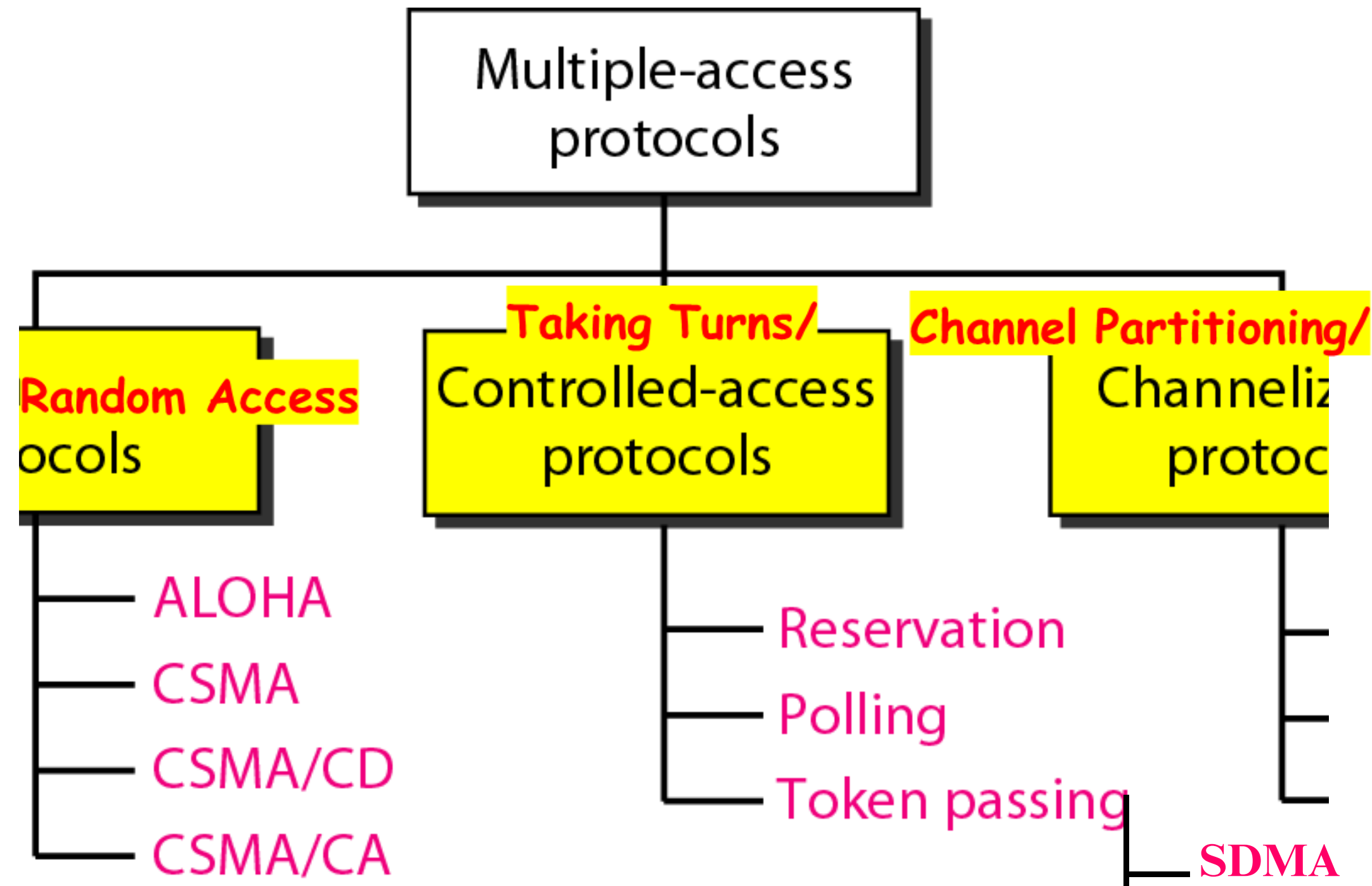
1. when one node wants to transmit, it can send at rate R .
2. when M nodes want to transmit, each can send at average rate R/M
3. fully decentralized:
 - no special node to coordinate transmissions
 - no synchronization of clocks, slots
4. simple

MAC Protocols: a taxonomy

Three broad classes:

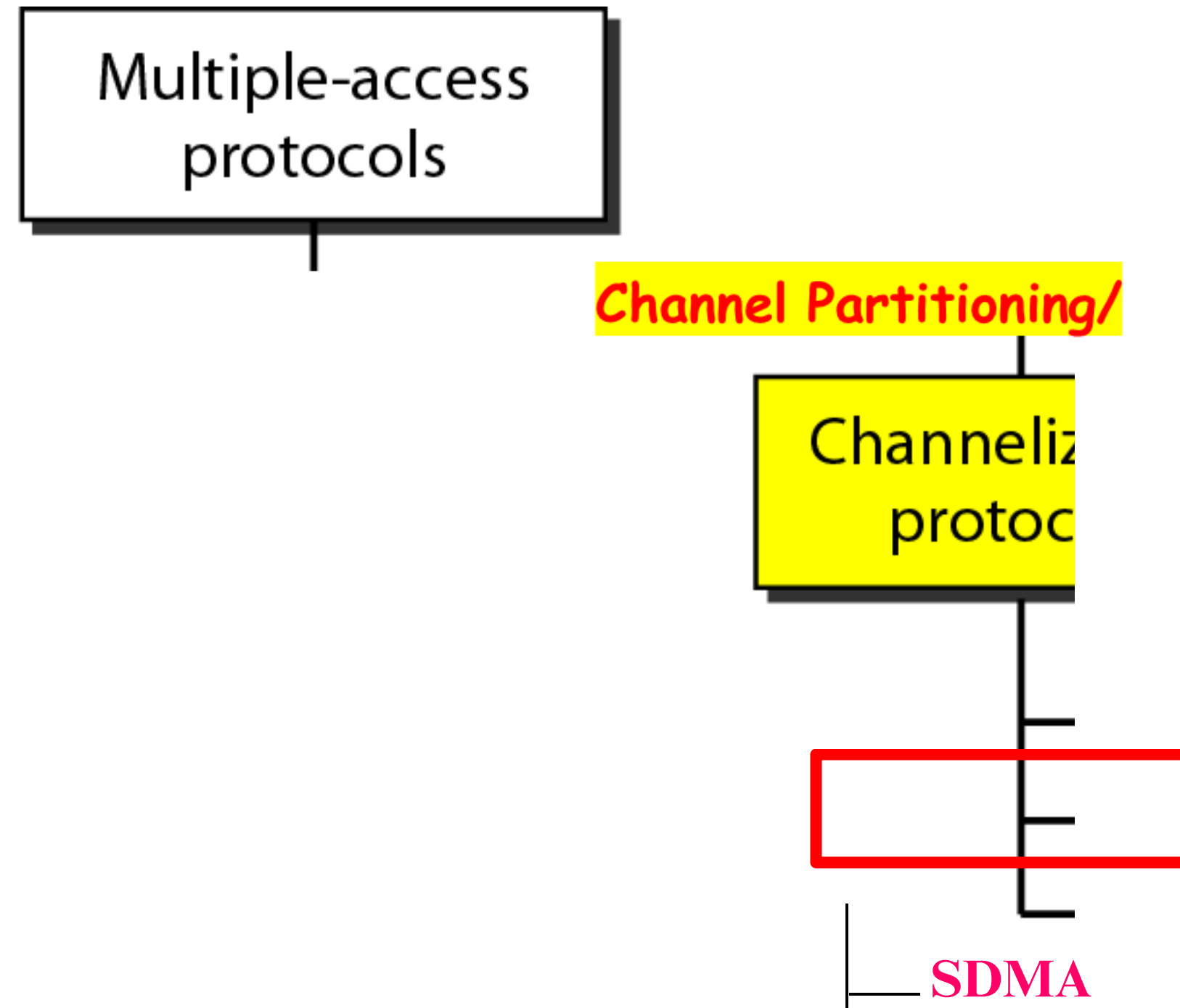
- **Channel Partitioning**
 - divide channel into smaller "pieces" (time slots, frequency, code)
 - allocate piece to node for exclusive use
- **Random Access**
 - channel not divided, allow collisions
 - "recover" from collisions
- **"Taking turns"**
 - nodes take turns, but nodes with more to send can take longer turns

Figure 1.2 *Taxonomy of multiple-access protocols discussed in this chapter*



Channel Partitioning

Dividing Time



Dividing time

Time-division multiple access (TDMA)

- access the channel in **rounds**
- Simplest scheme: each node gets fixed length **slot**
 - length of the slot = packet transmission time
- unused slots go idle
- example: 6-node link, 1,3,4 have packets to send, slots 2,5,6 idle

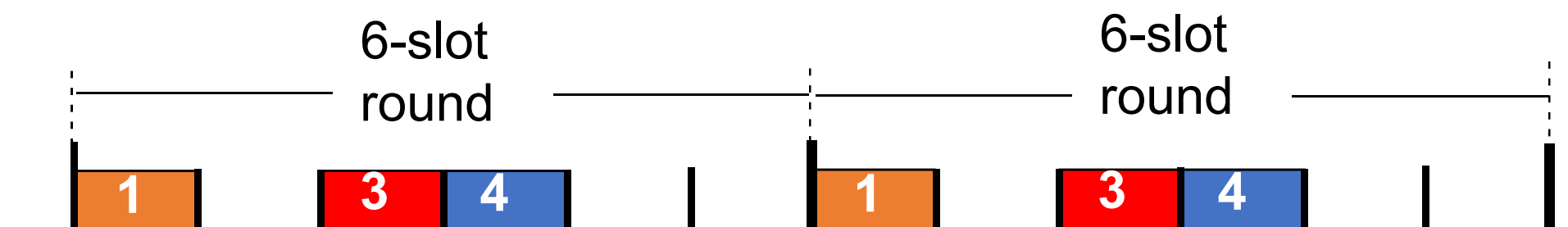
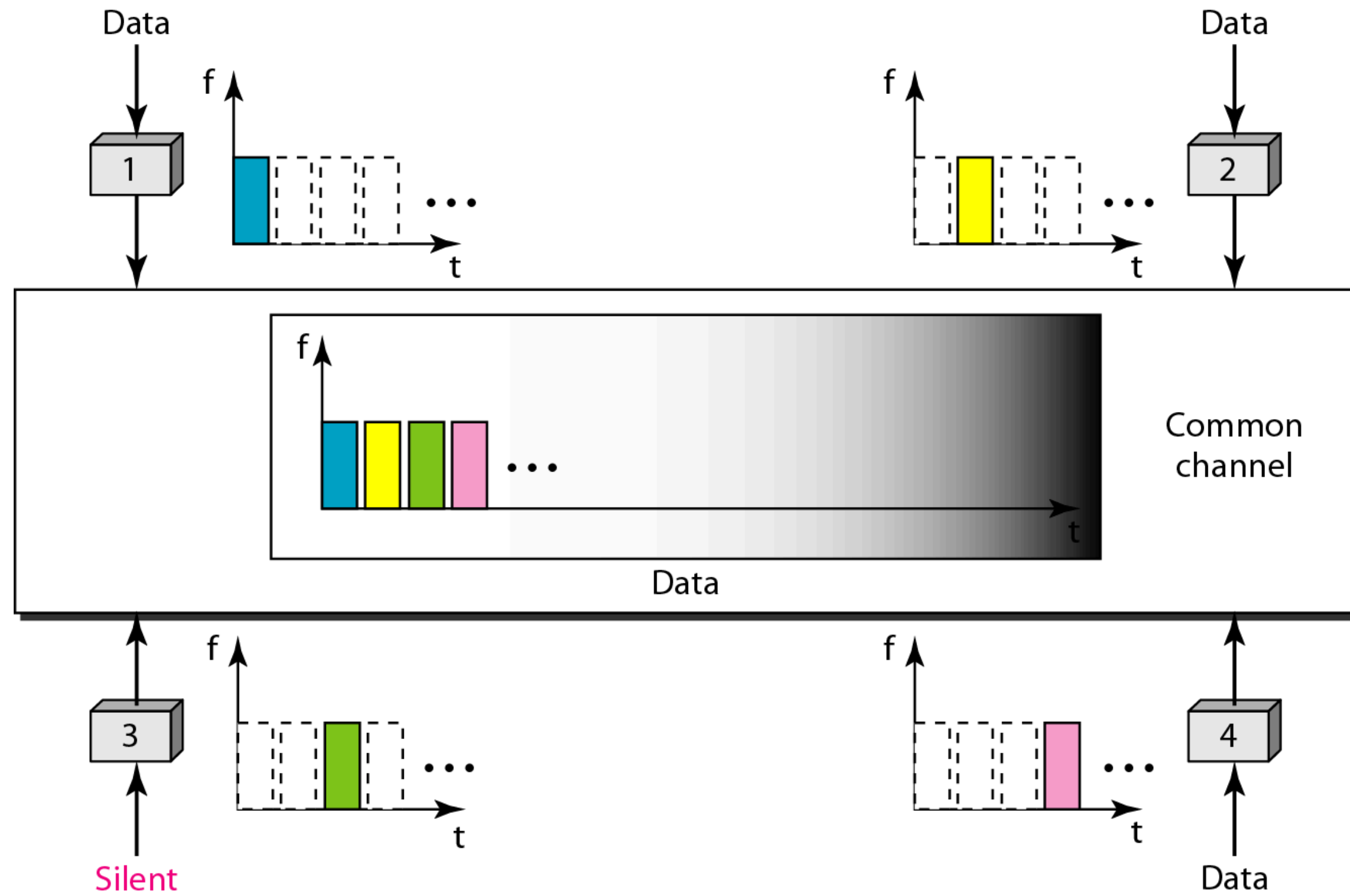
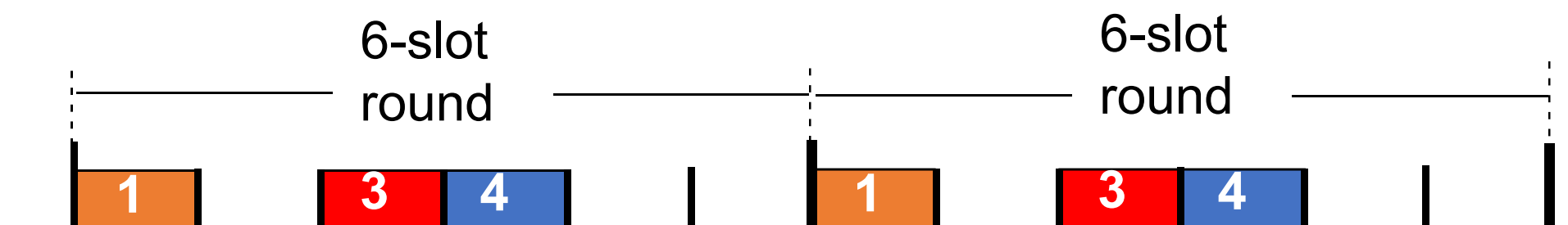


Figure 12.22 *Time-division multiple access (TDMA)*

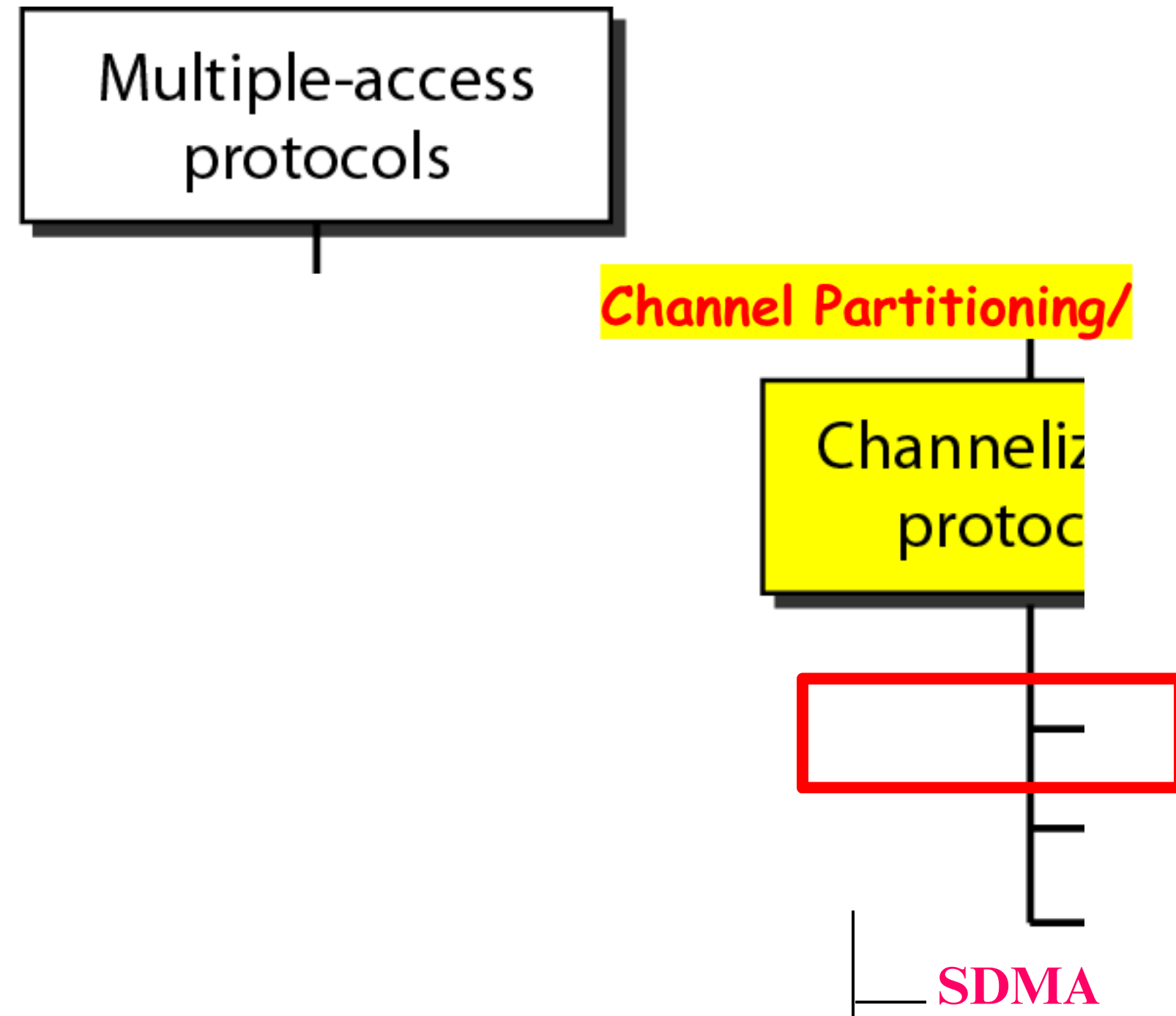


Dividing time

- If channel is partitioned statically, users with higher demand cannot use idle slots. **Inefficient**
- Example: Nodes 1, 3, or 4 may have more data to send, yet cannot use the slots of nodes 2, 5, or 6.



Dividing Frequency

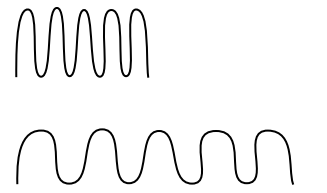


Dividing frequency

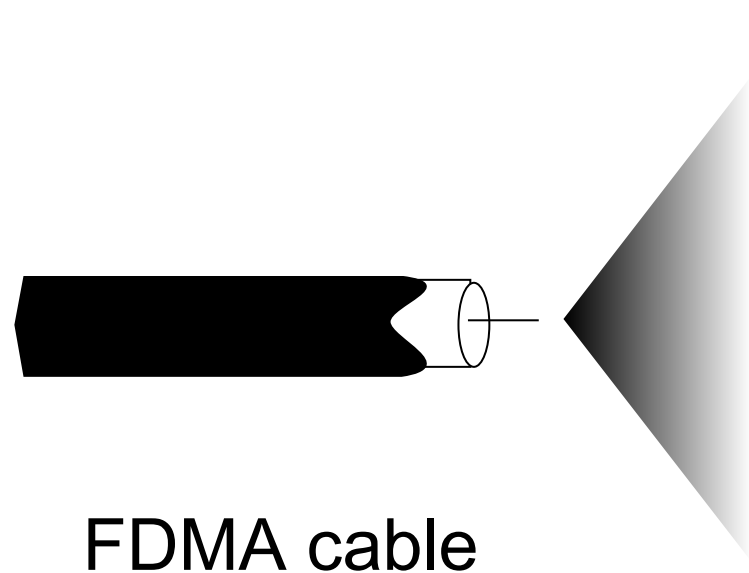
FDMA: frequency division multiple access

- Channel divided into **frequency bands**
- Simple scheme: each node assigned a fixed frequency band
- Unused transmission time in frequency bands go idle
- Example: 6-node link, 1,3,4 have packets to send, frequency bands 2,5,6 idle

High frequency



Low frequency



FDMA cable

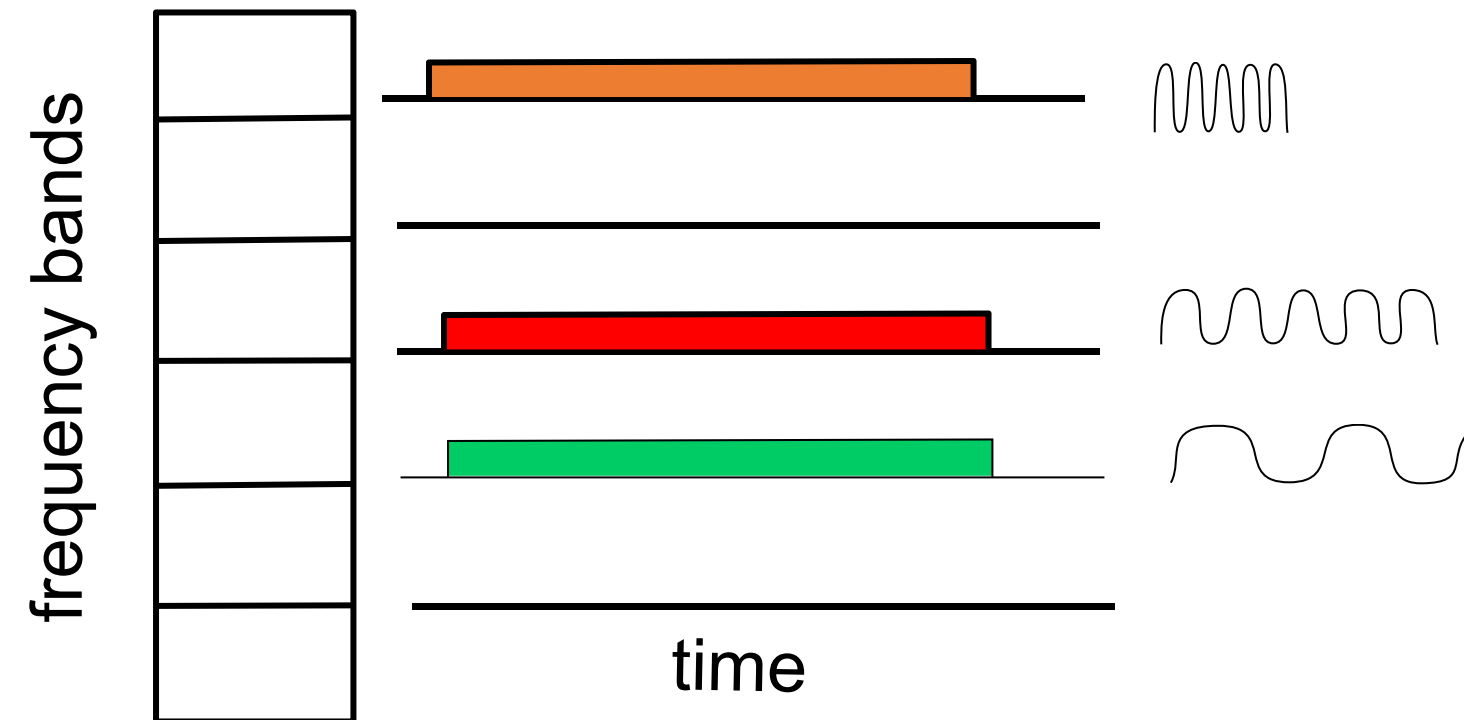
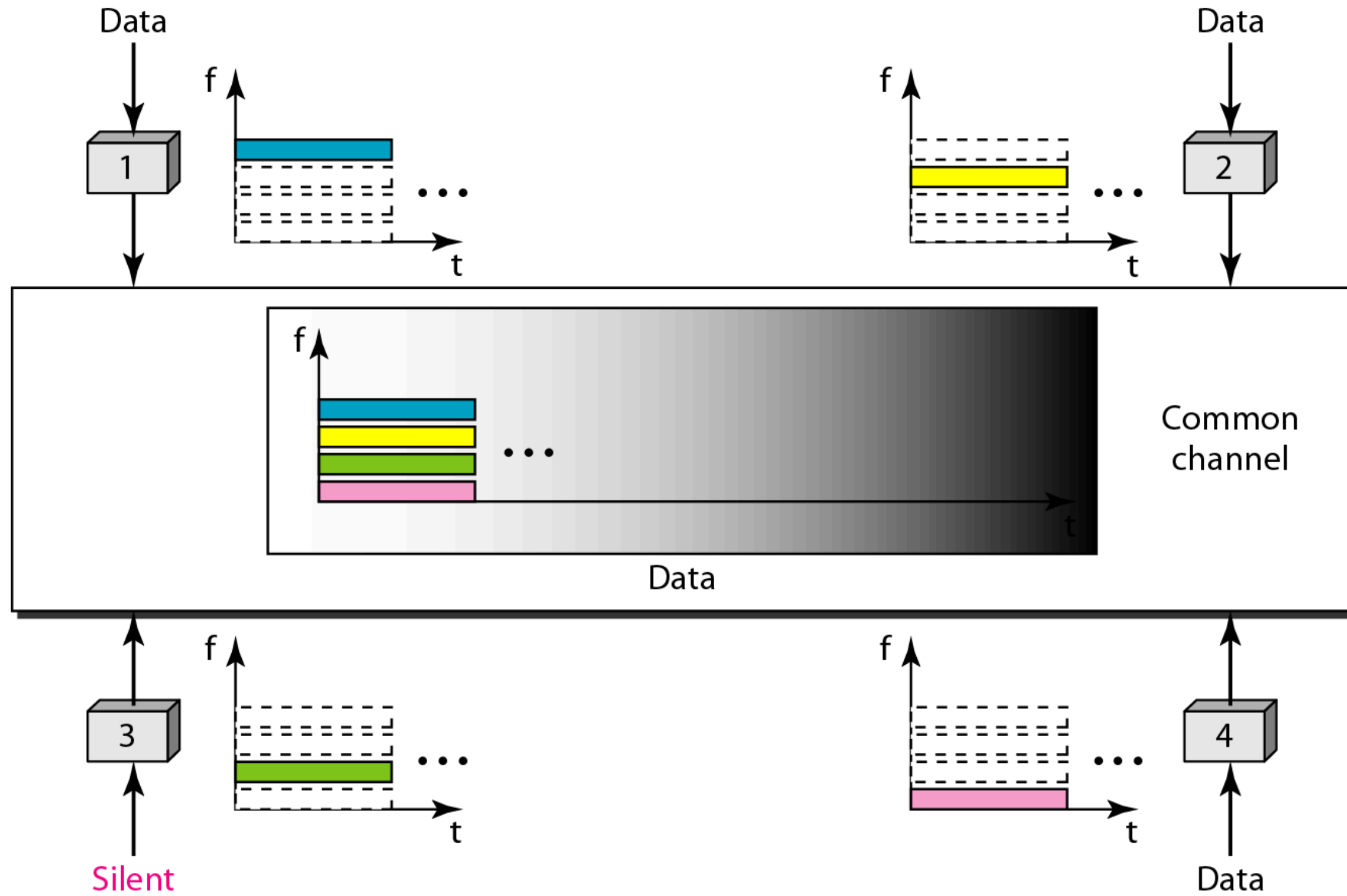
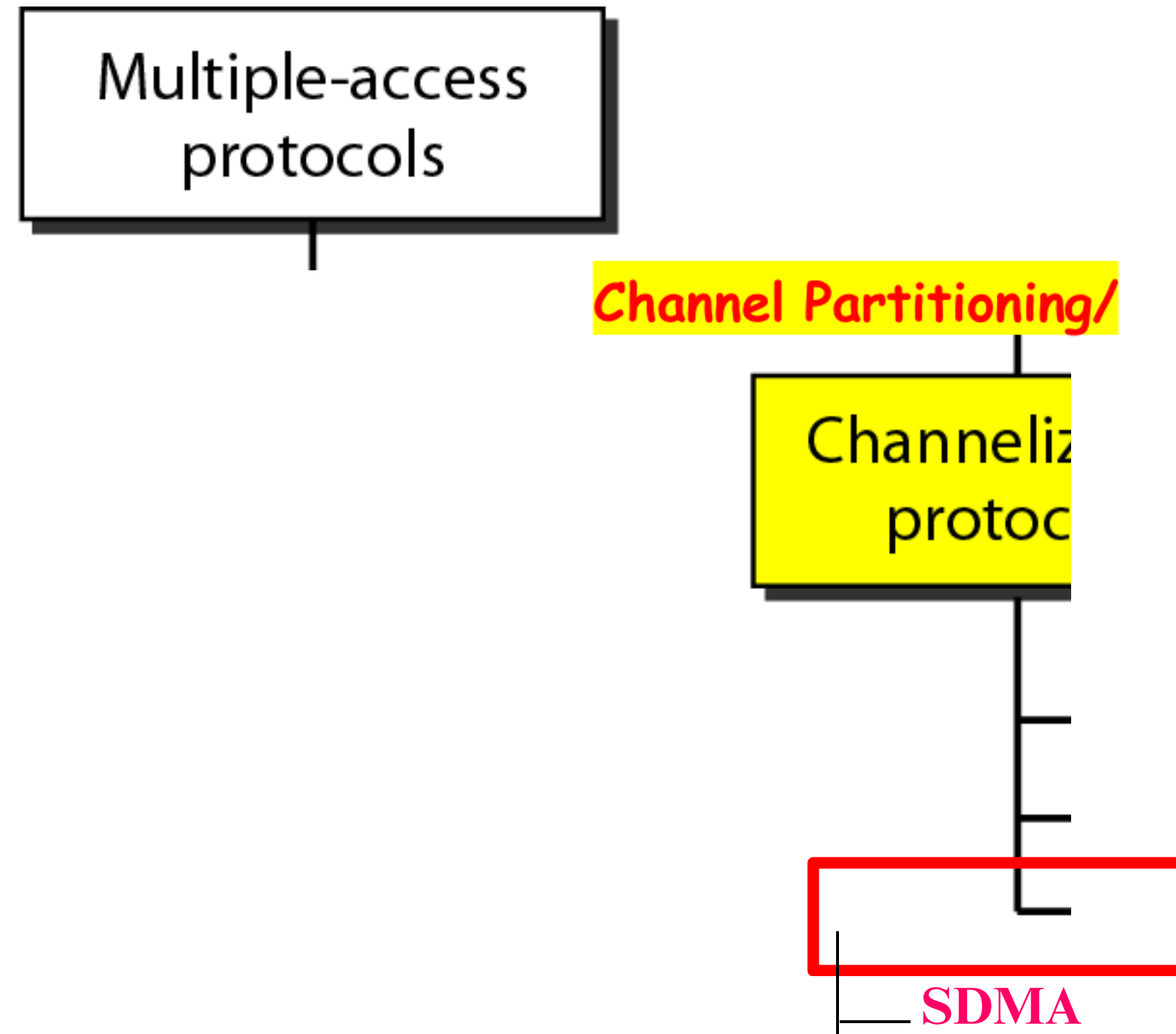


Figure 12.21 *Frequency-division multiple access (FDMA)*



Dividing Code





Note

In CDMA, one channel carries all transmissions simultaneously.

Dividing code

Code Division Multiple Access (CDMA)

- A unique **code** is assigned to each pair of nodes
 - Also called chipping sequence
- Users can **share frequency bands** and can **transmit at the same time**
- Encoded signal = data signal * chipping sequence
- Decoded signal = received signal X chipping sequence
 - Here “X” denotes the inner product of two vectors
 - Example: $[1, 2, 3] \times [4, 5, 6] = [1*4, 2*5, 3*6]/3 = [4/3, 10/3, 18/3]$

Figure 12.23 *Simple idea of communication with code*

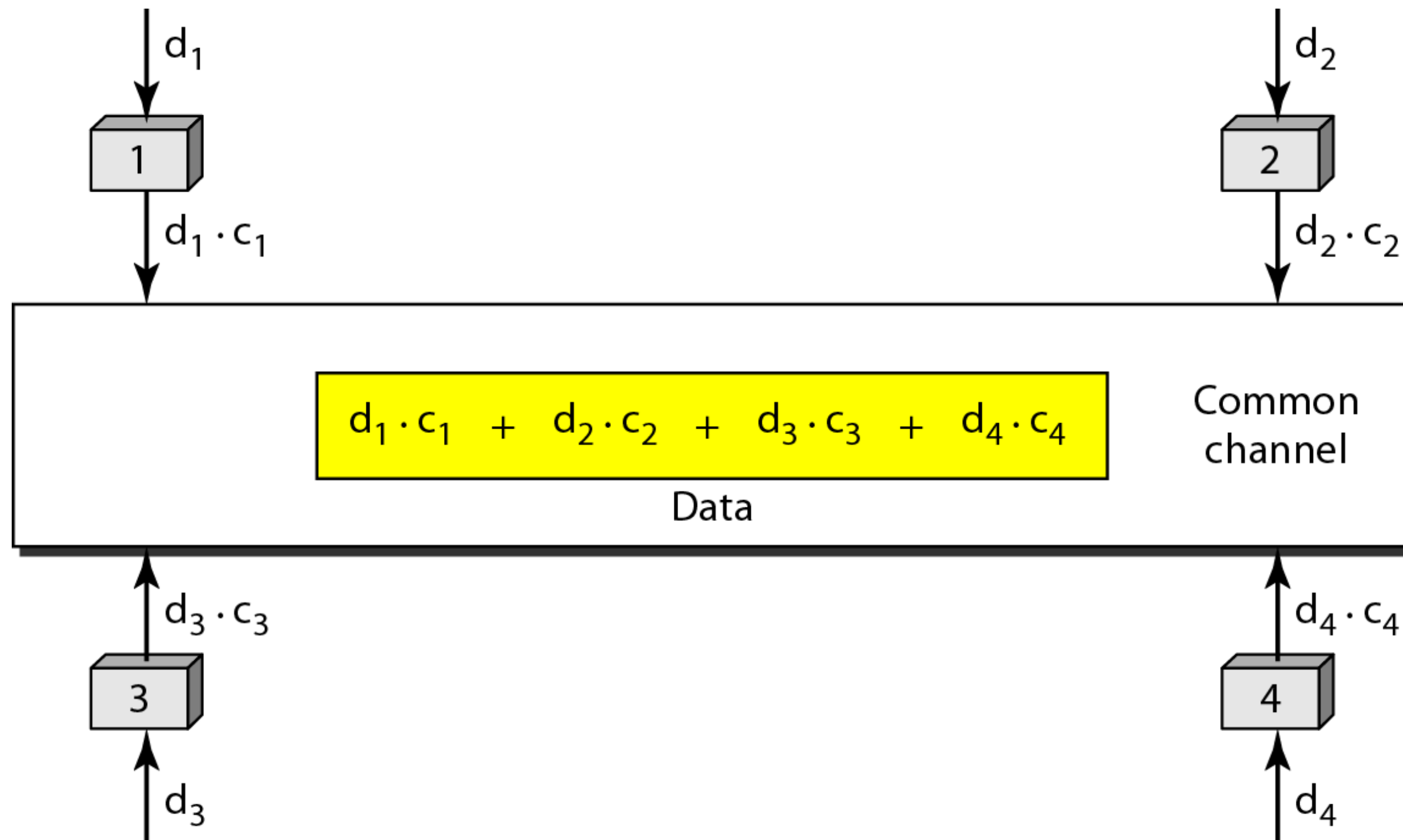


Figure 12.24 *Chip
sequences*

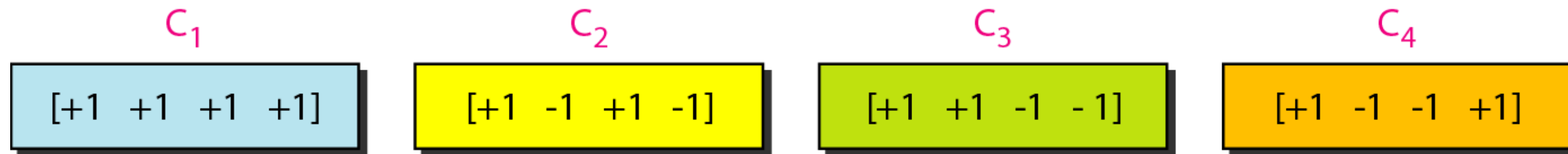


Figure 12.25 *Data representation in CDMA*

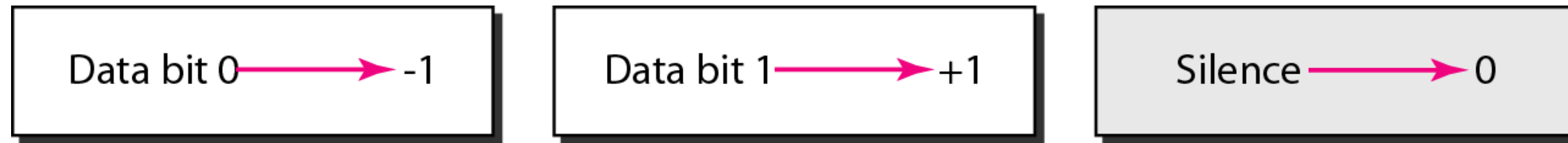


Figure 12.26 *Sharing channel in CDMA*

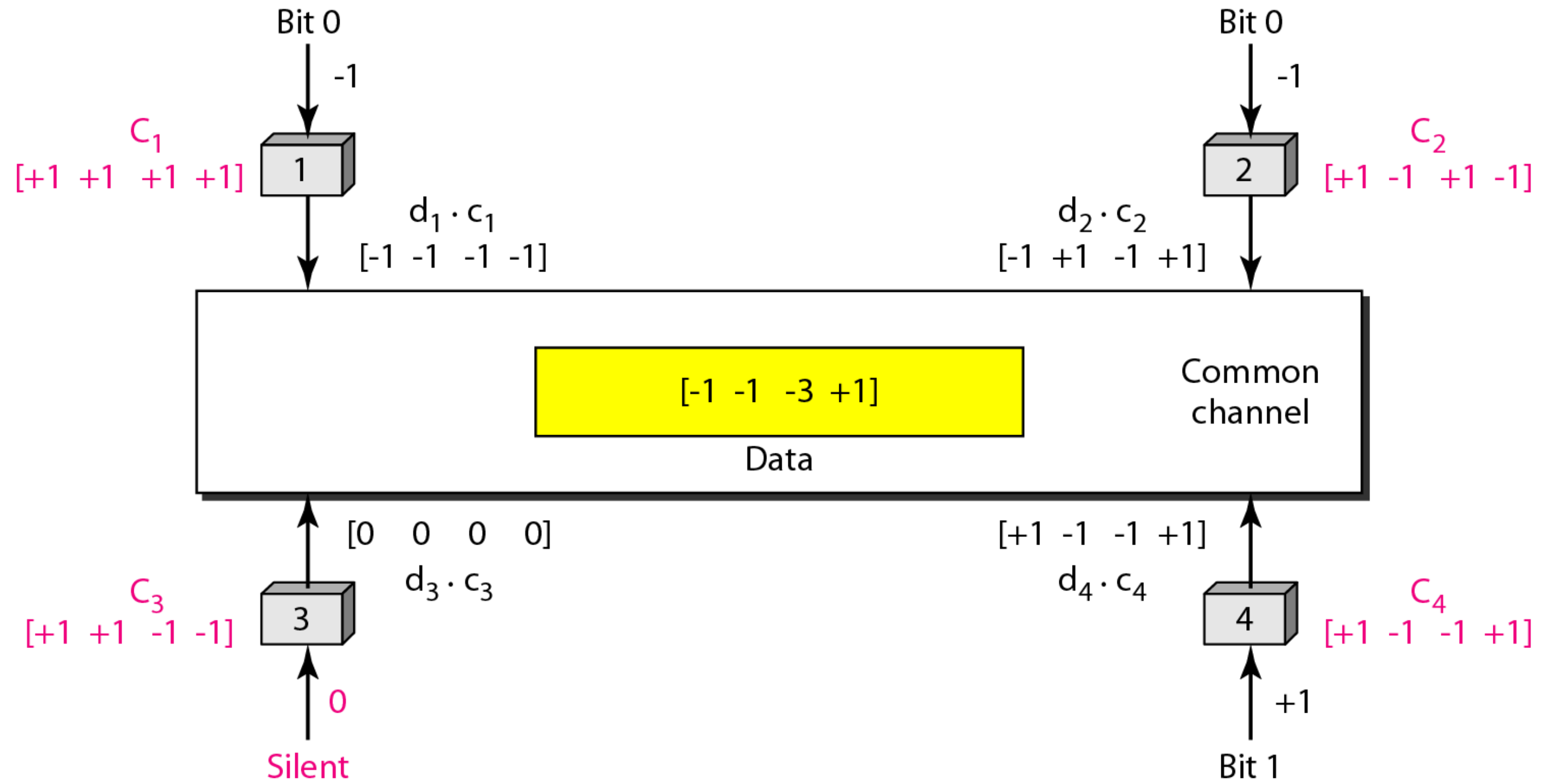


Figure 12.27 *Digital signal created by four stations in CDMA*

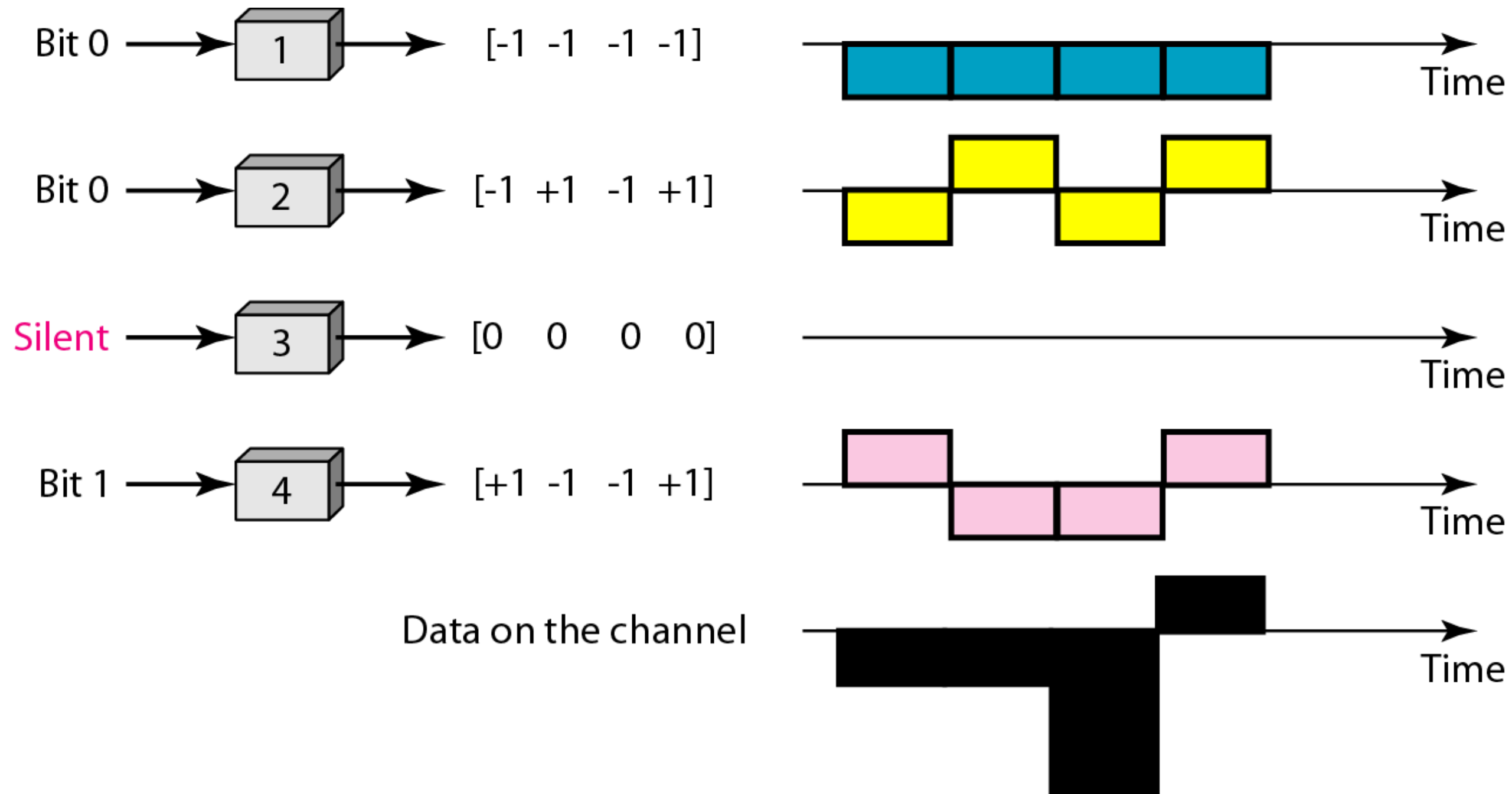
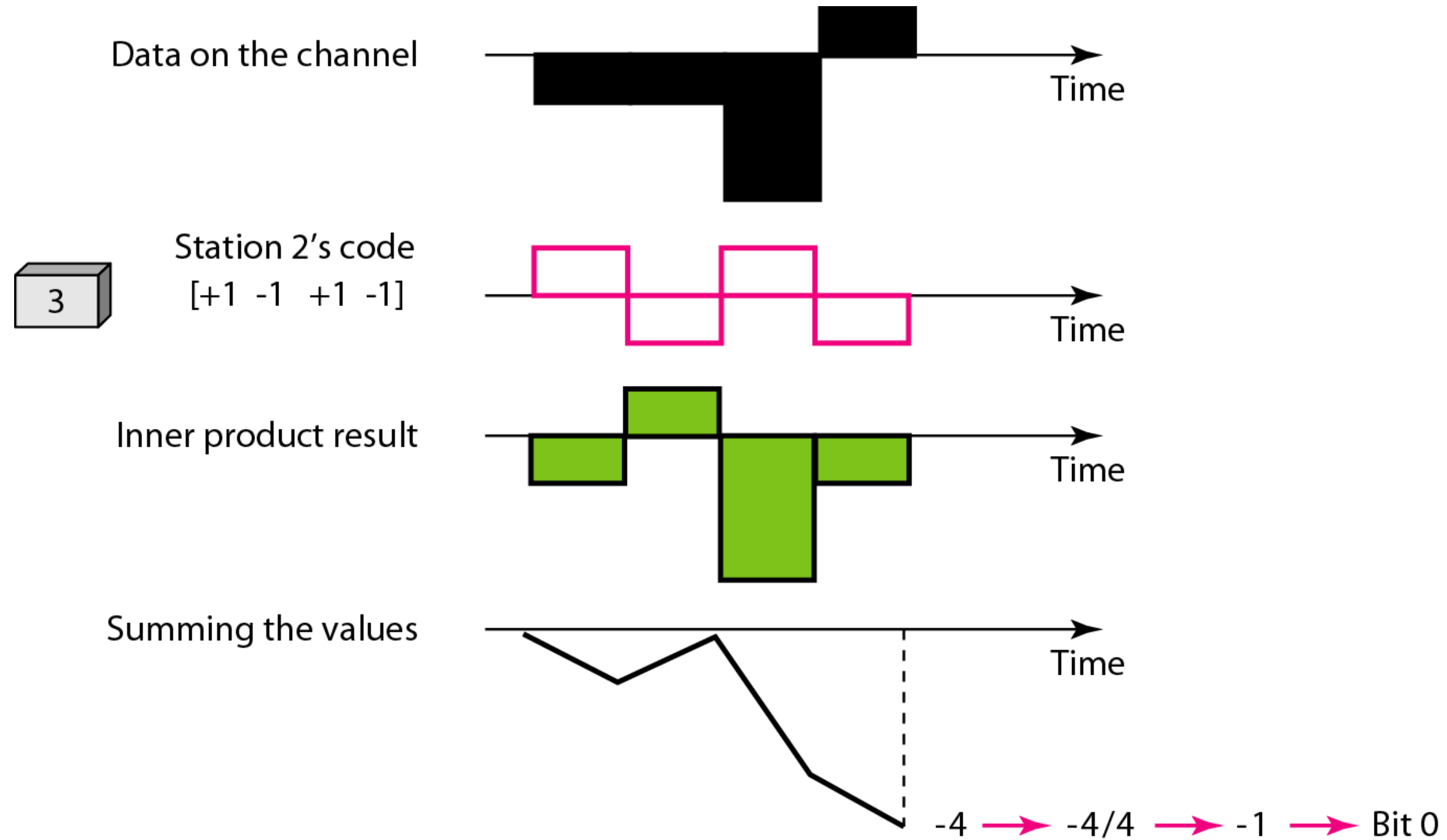
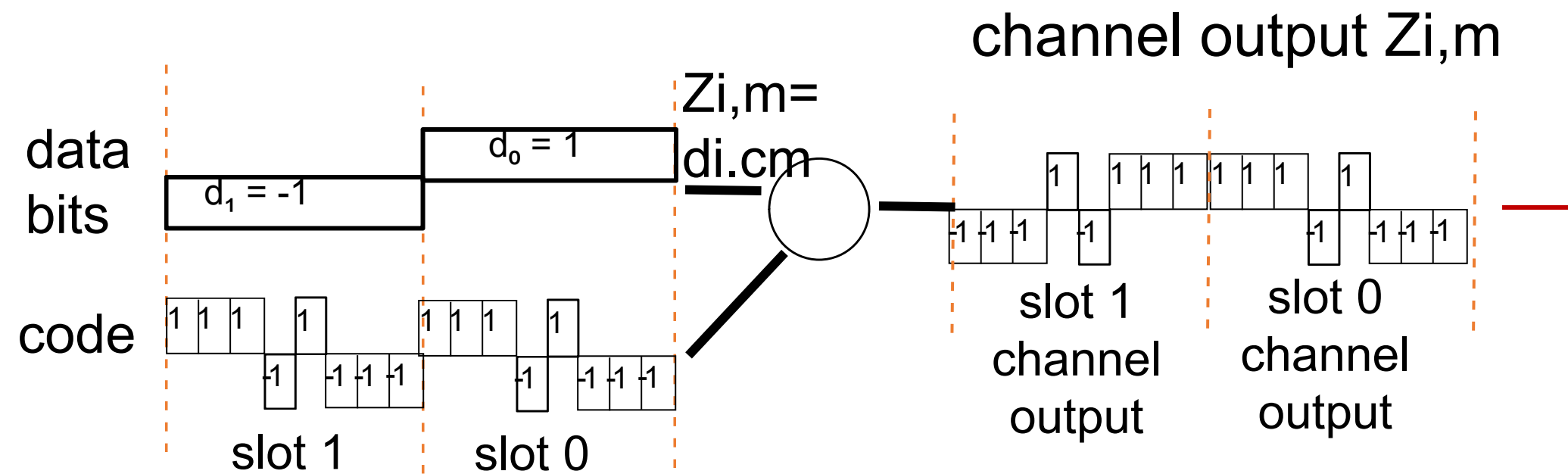


Figure 12.28 *Decoding of the composite signal for one in CDMA*

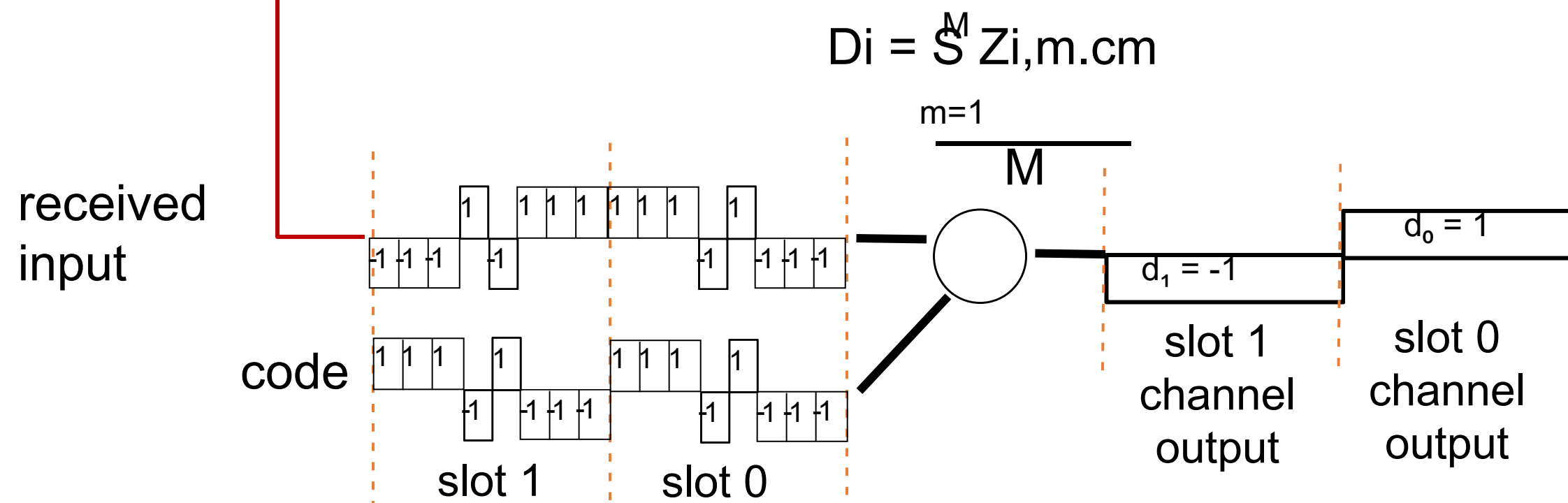


Example of CDMA encode/decode

sender



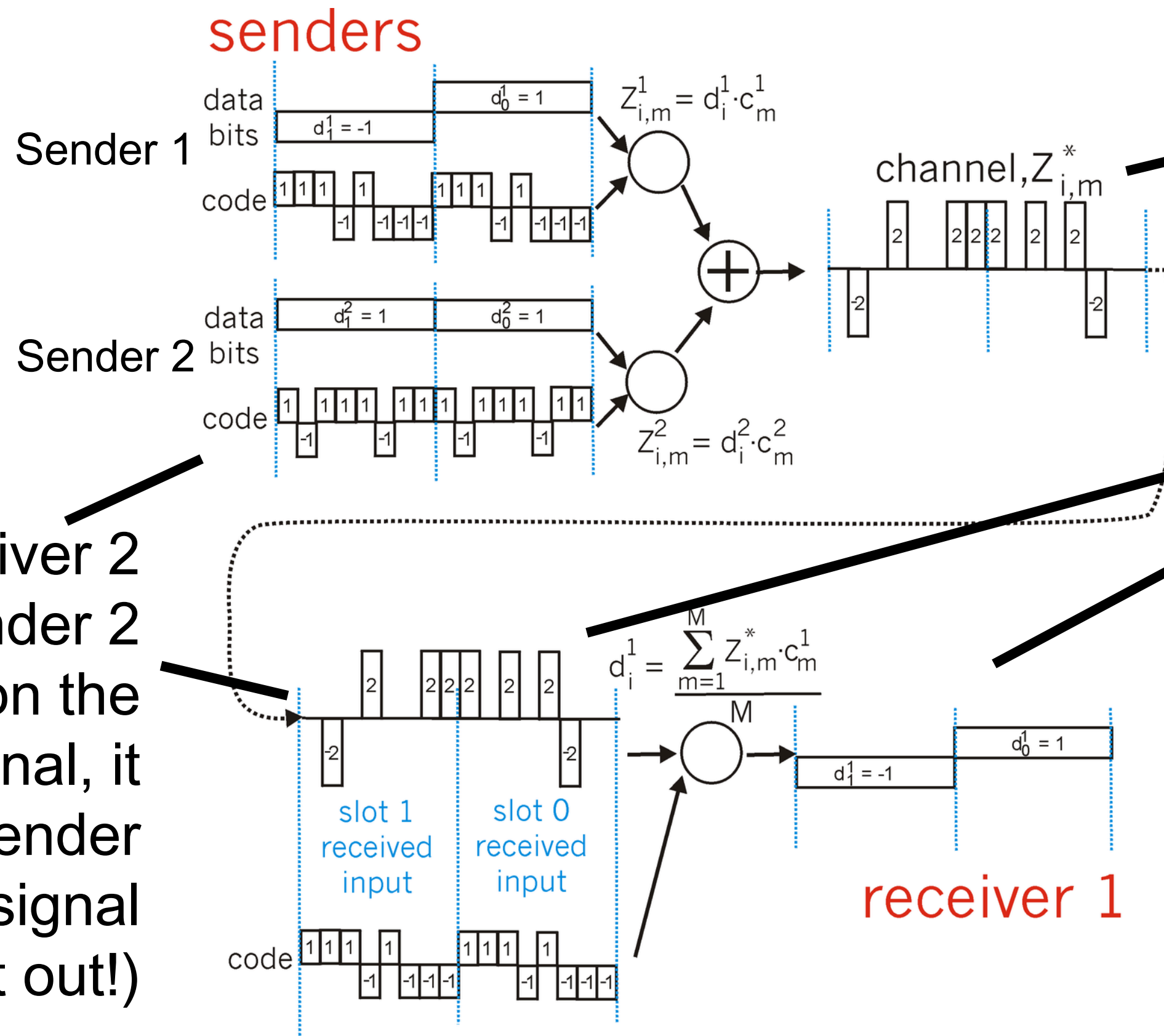
receiver



CDMA trade-offs

- In the place of transmitting 2 (data) bits, we're sending $2 * 8 = 16$ bits on the channel. Seems wasteful?
- Real benefit: when **multiple users** transmit
- Suppose resulting signal levels just add up in the channel and the sum is received by each receiver
- With CDMA, **each receiver can independently decode its corresponding sender's transmission** using the (respective) code!

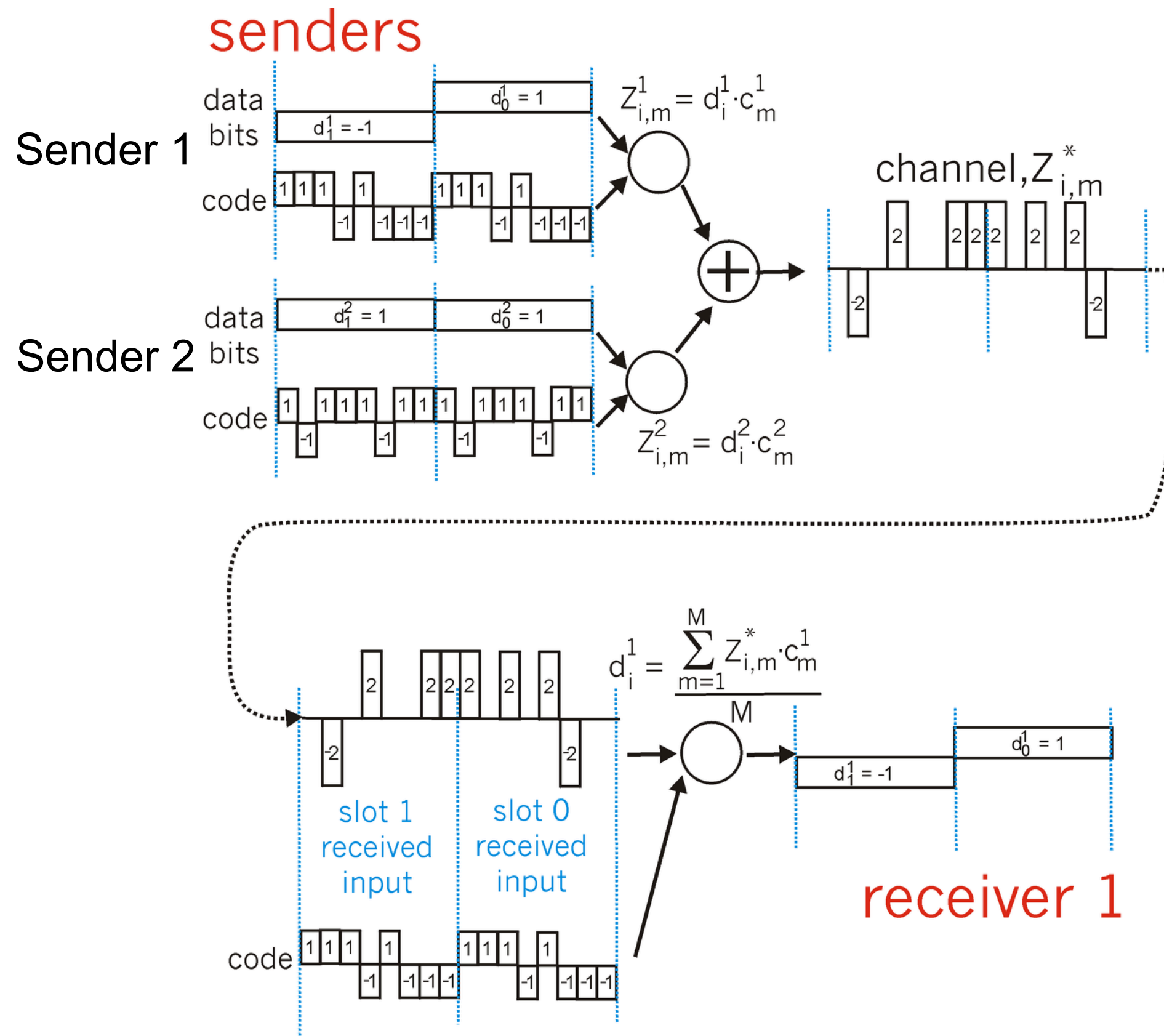
CDMA with two senders



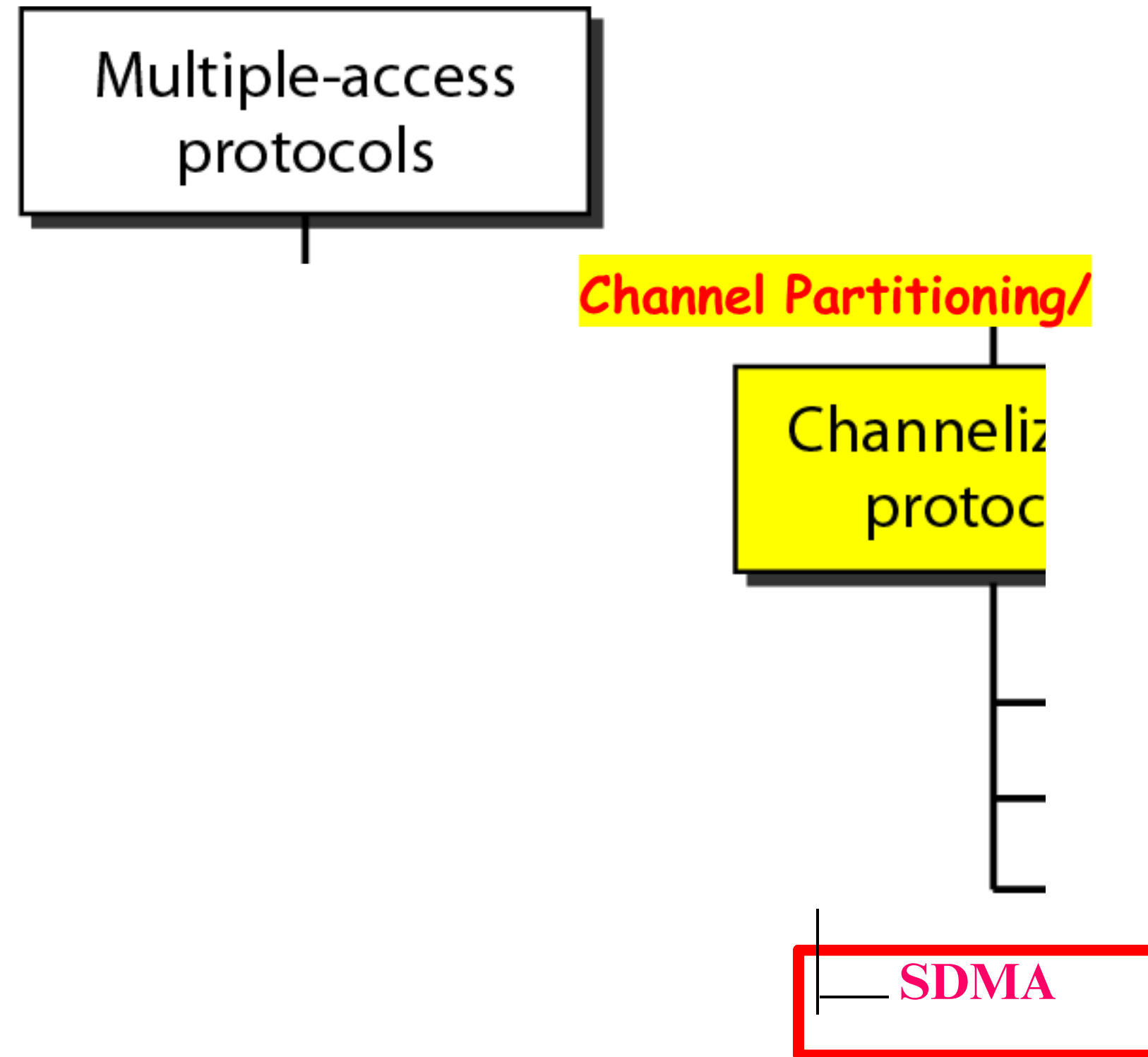
channel sums together transmissions by sender 1 and 2

using same code as sender 1, receiver recovers sender 1's transmission from summed channel data!

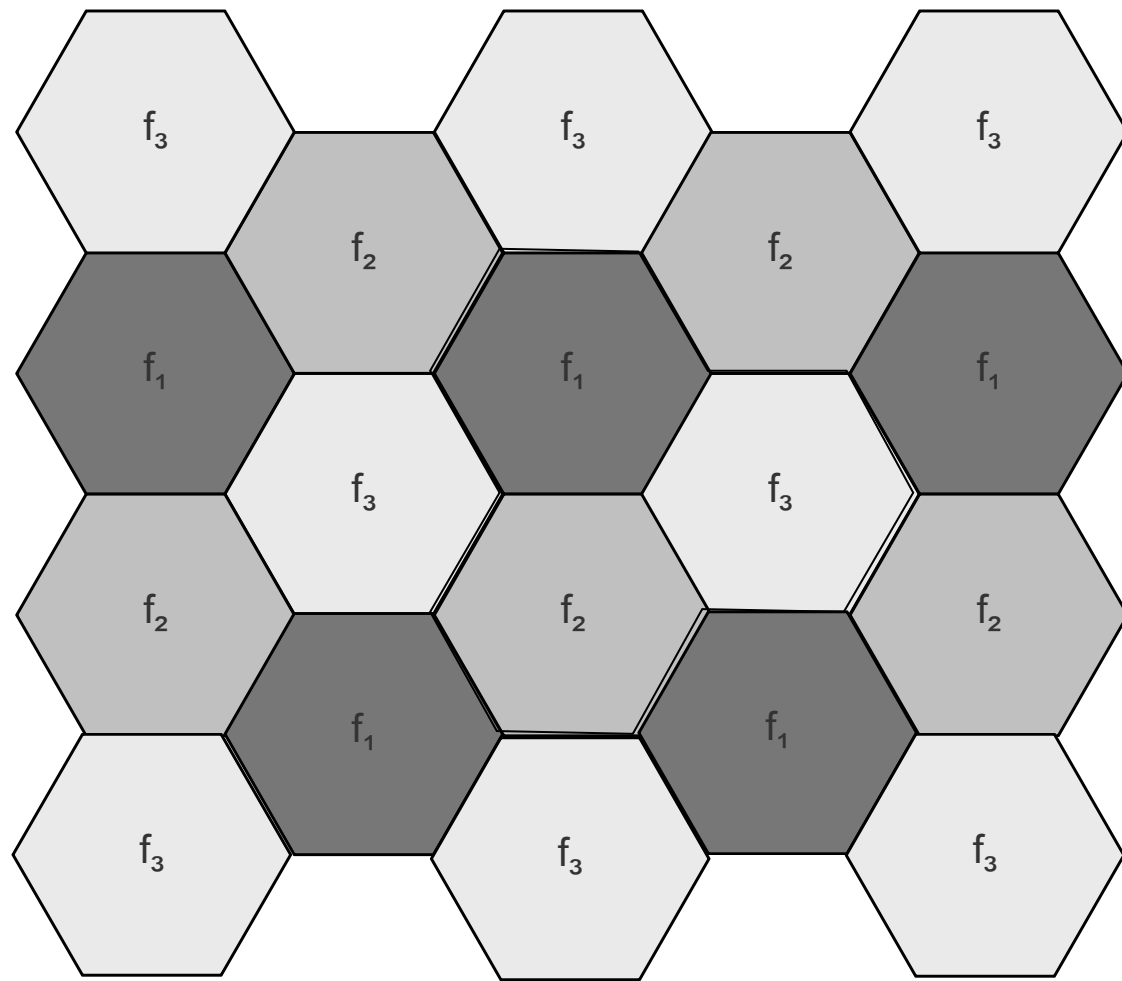
CDMA with two senders



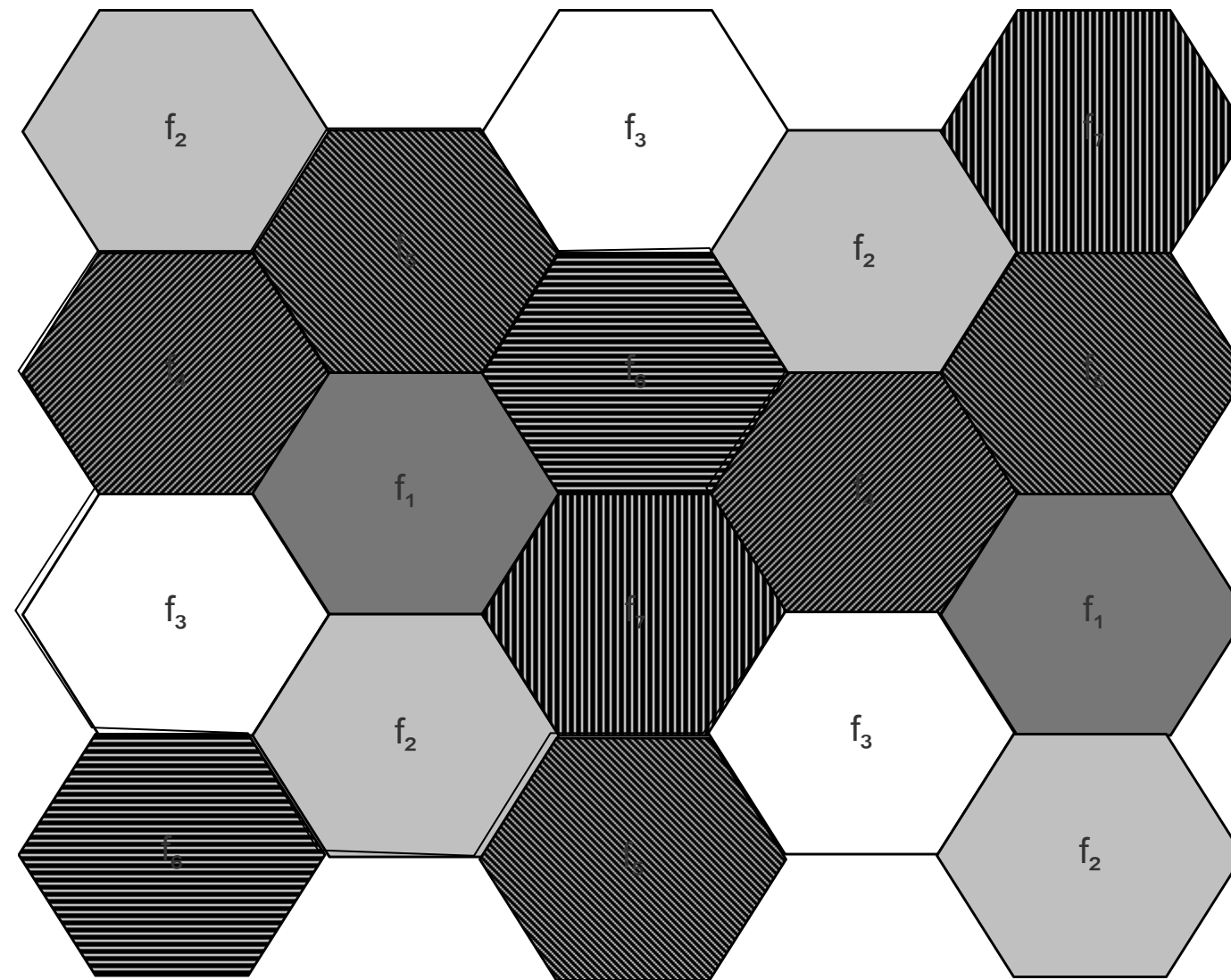
Dividing Code



Dividing Space SDMA

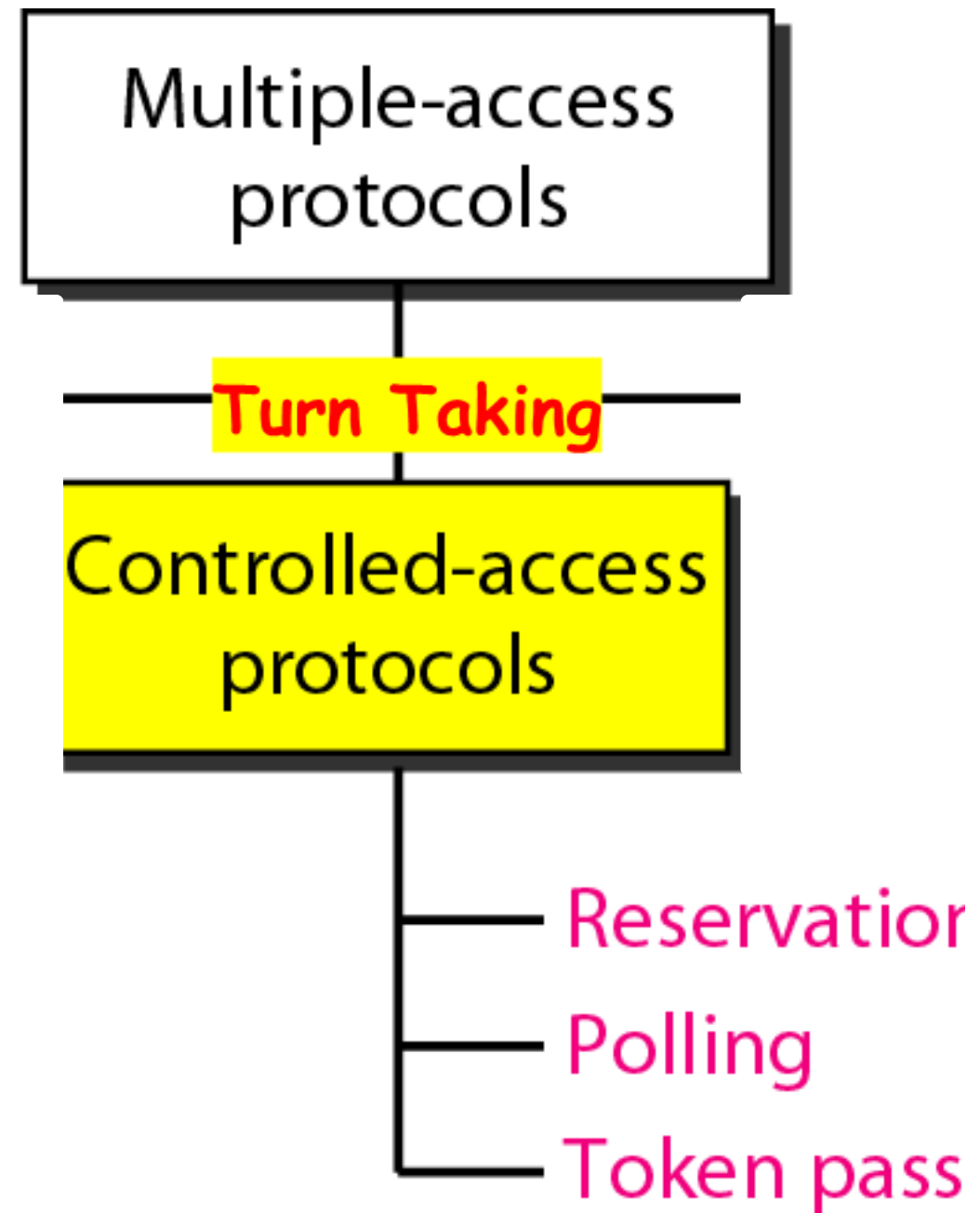


3 cell cluster



7 cell cluster

Turn-taking protocols



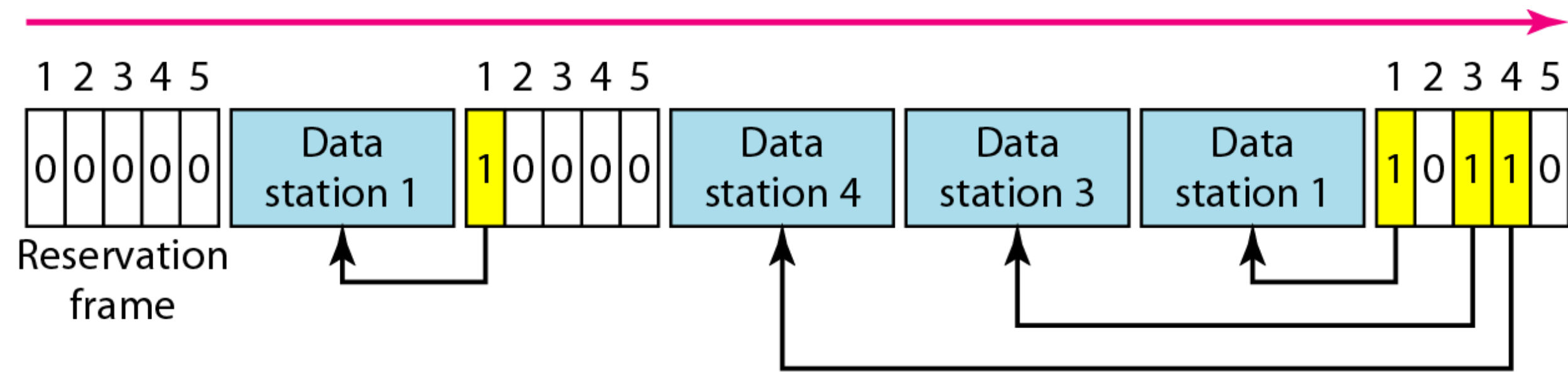
Turn-taking protocols

In **Turn-taking protocols**, the stations consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations. We discuss three popular controlled-access methods.

- Reservation
- Polling
- Token Passing

Reservation access method

Figure 12.18 Reservation access method



Polling

- Orchestrator node “invites” sender nodes to transmit in turn
- Concerns:
 - polling overhead
 - latency
 - single point of failure
 - The orchestrator

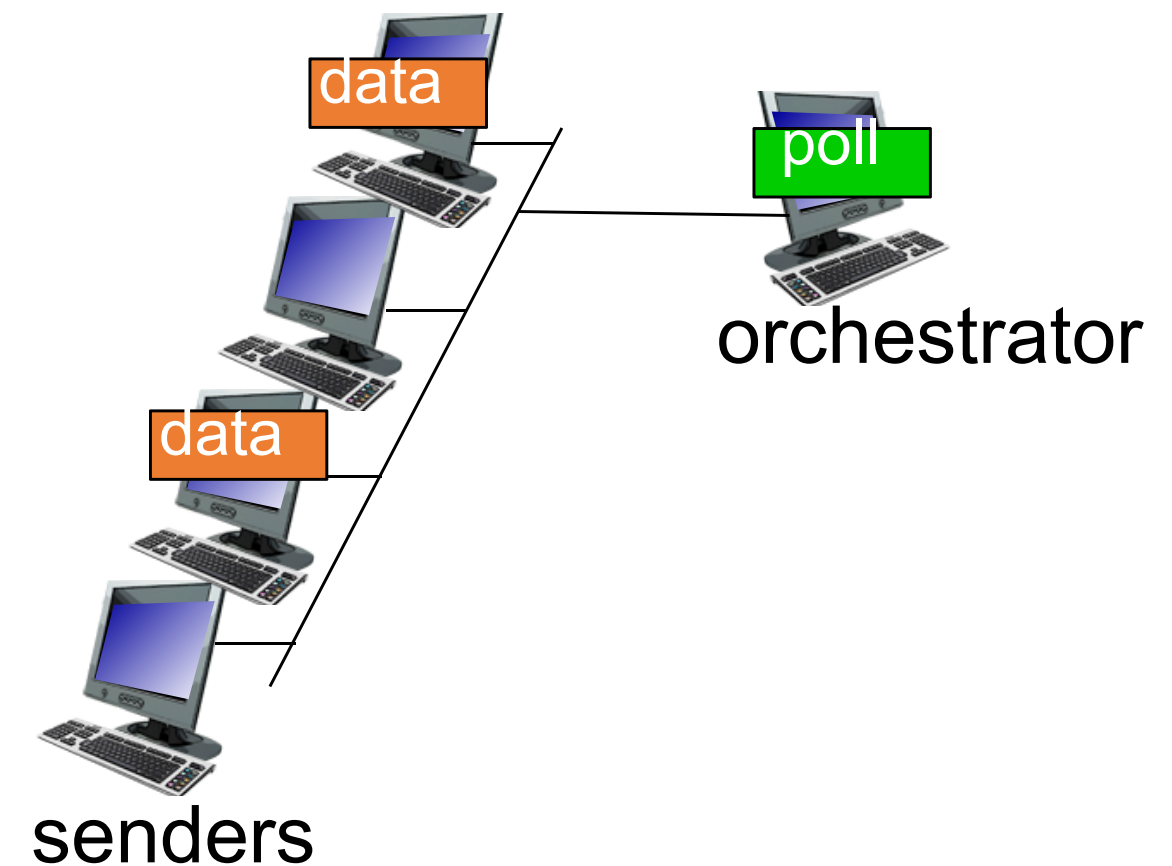
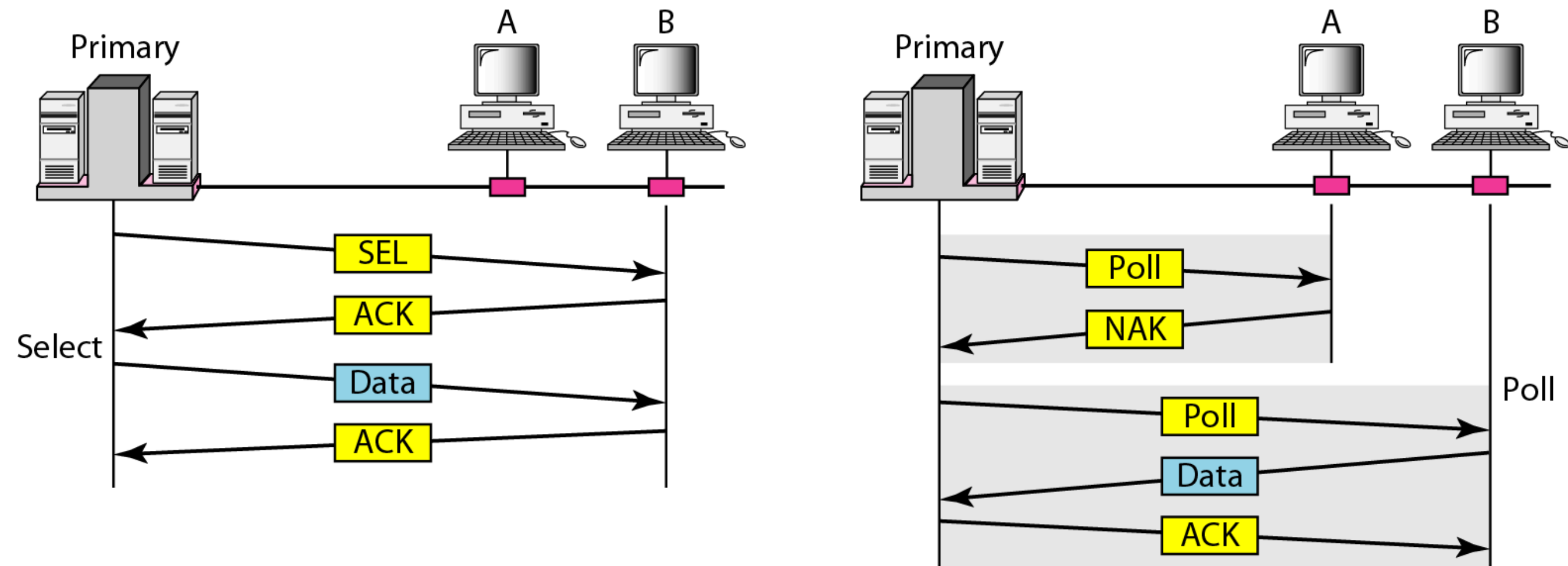


Figure 12.19 *Select and poll functions in polling access method*



Token passing

- control token passed from one node to next sequentially
- Node with token can transmit
- Else, pass it on to the next node in the “ring”
- Concerns:
 - latency
 - single point of failure (node holding the token)

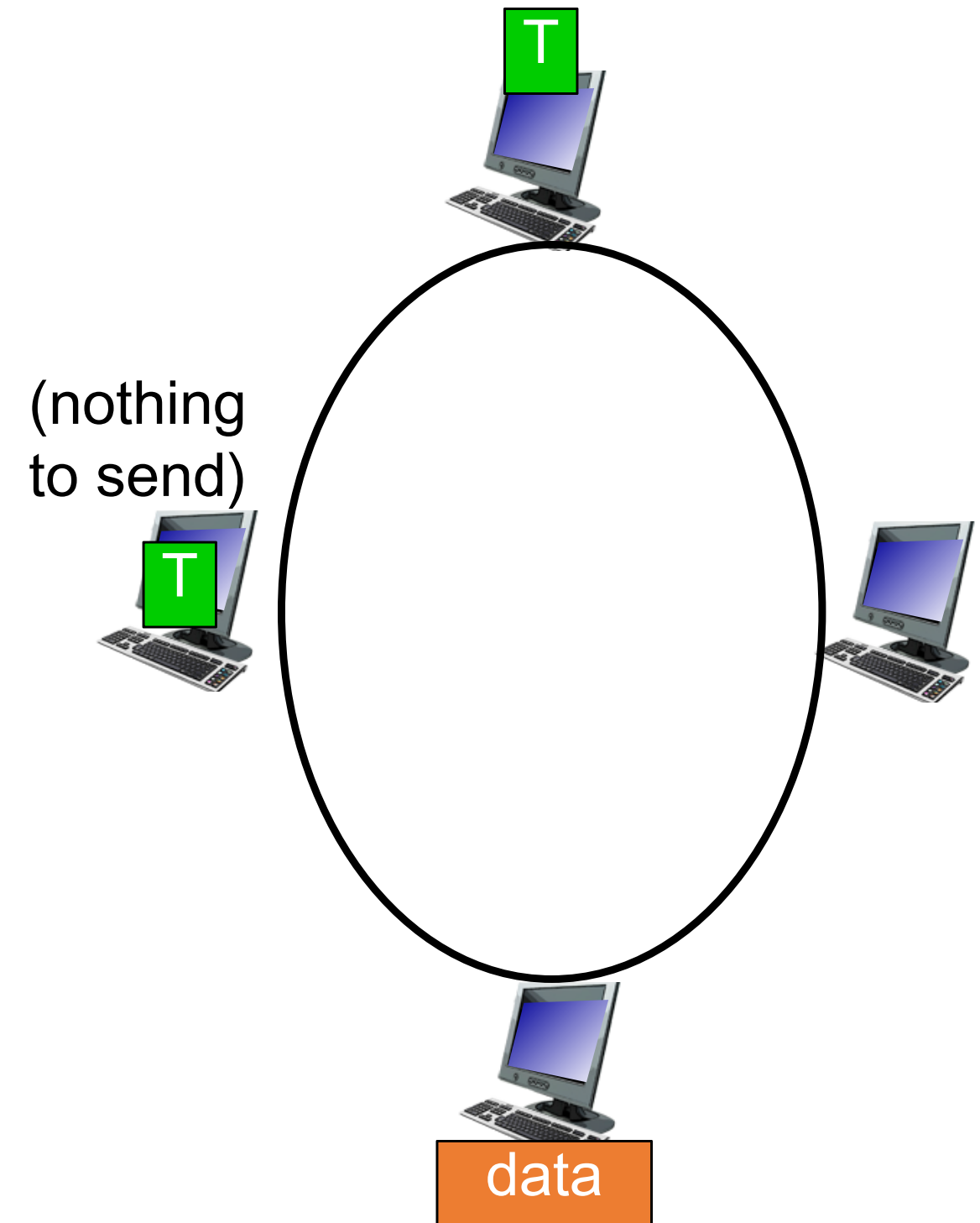
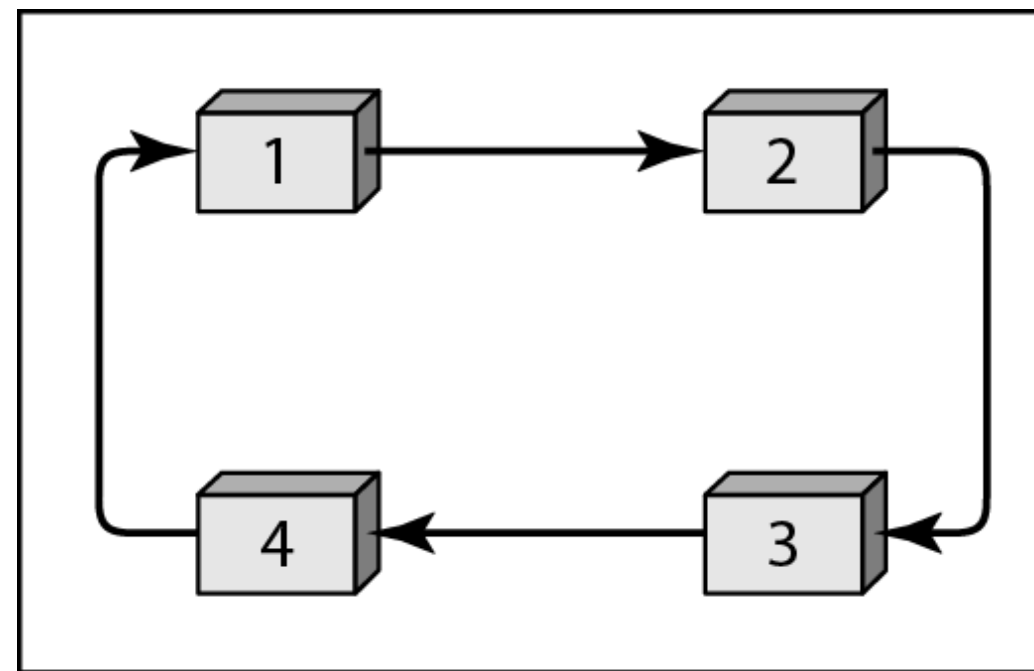
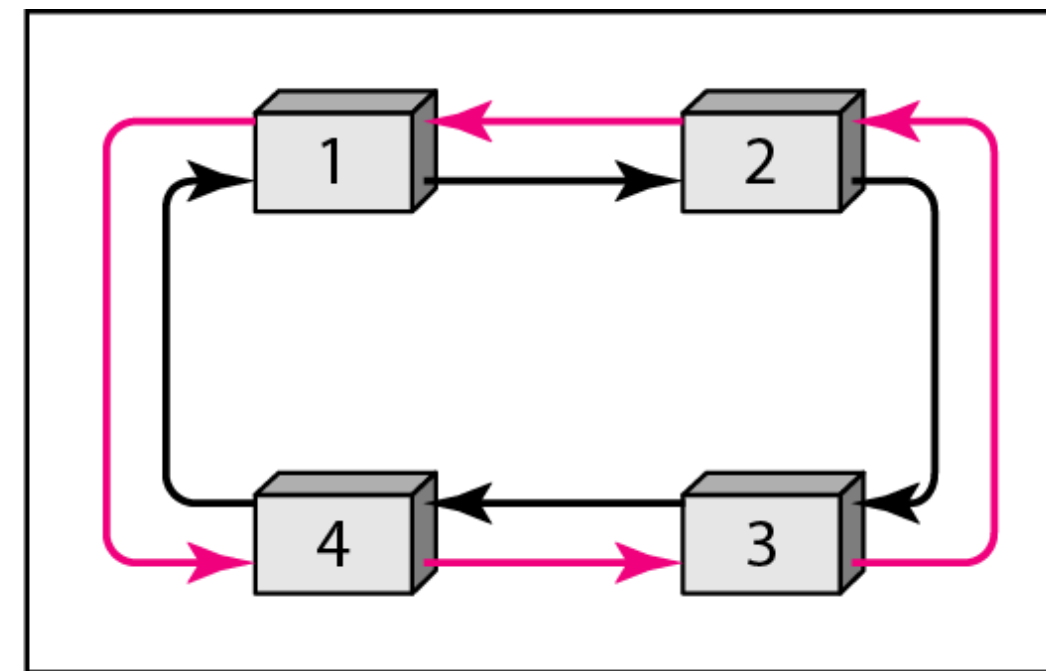


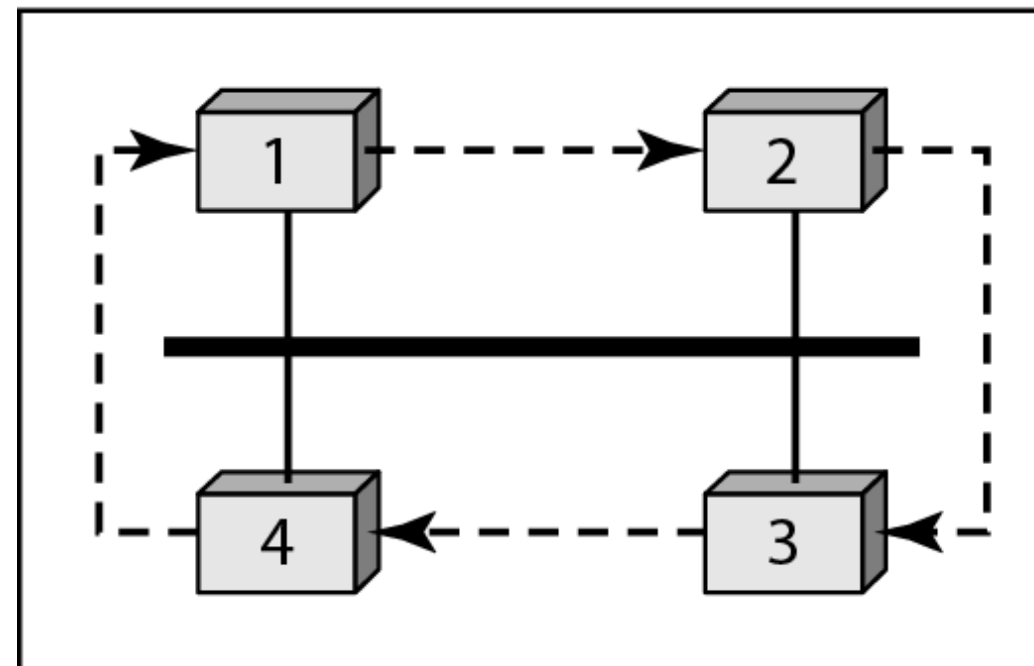
Figure 12.20 *Logical ring and physical topology in token-passing access method*



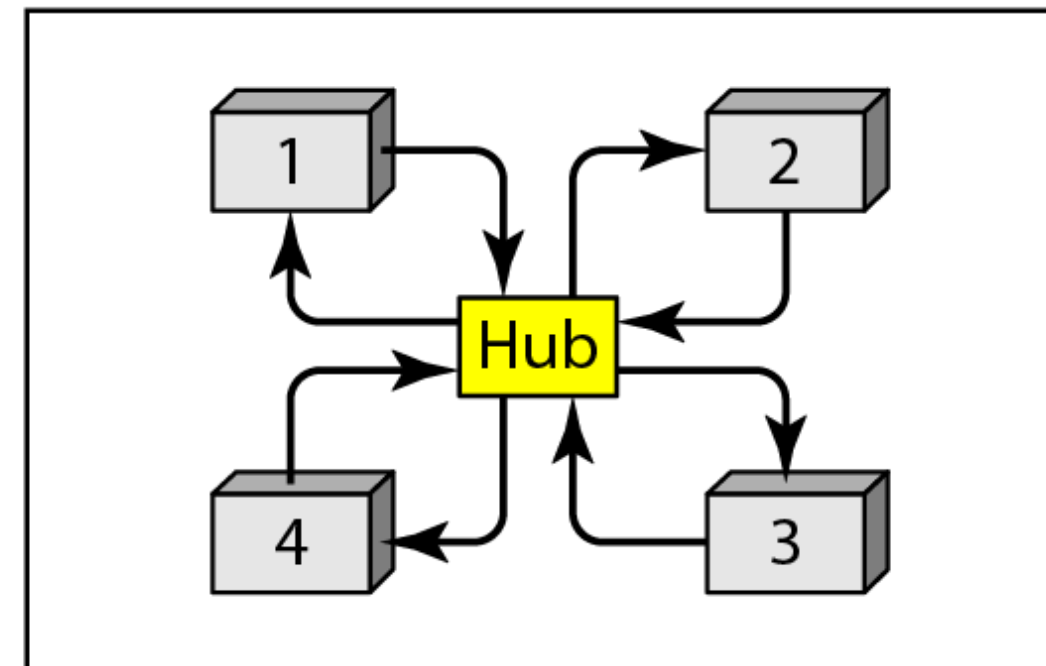
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

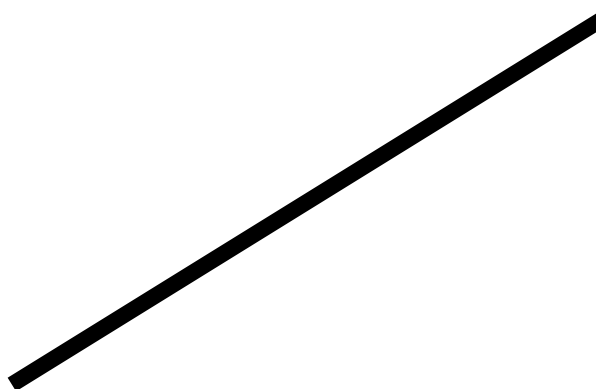
Summary

- Medium access control: a **distributed algorithm** to transmit over a shared link
- **Channel partitioning**: TDMA, FDMA, CDMA
- **Turn-taking**: polling, token passing
- TDMA, FDMA, & CDMA are widely used in cellular communication
 - TDMA, FDMA: GSM
 - CDMA: Qualcomm/Verizon

Review

Medium access control

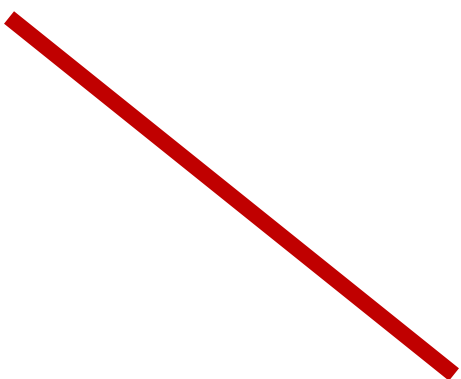
A distributed algorithm running at nodes to determine who should transmit over a shared link



Channel Partitioning

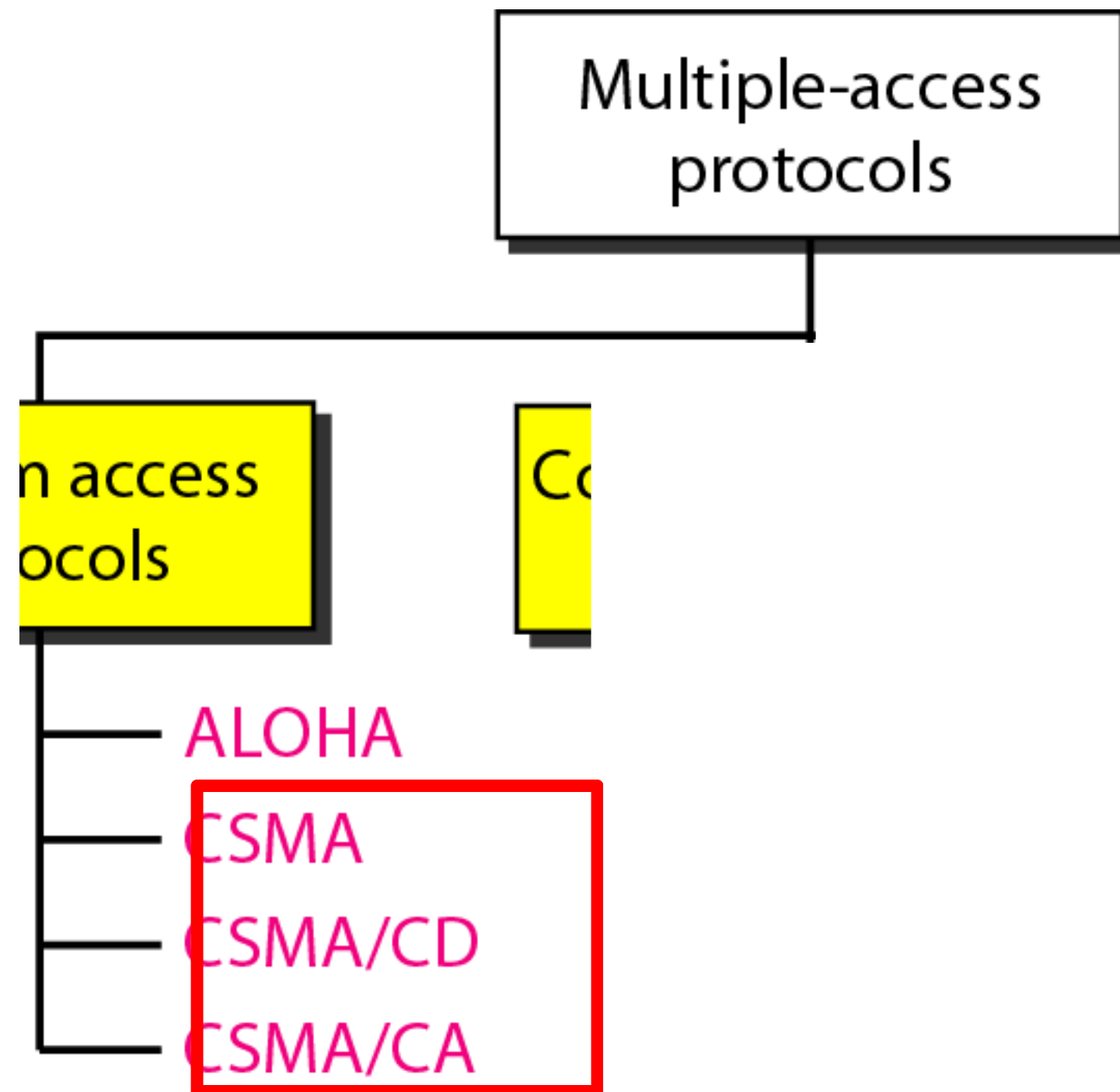


Turn taking



Random access

Random access



Random access protocols

Random access protocols

- What if nodes decide to transmit randomly?
- When node has packet to send
 - Transmit at **full channel data rate R**
 - No *a priori* coordination among nodes
- **Collision** possible when two or more transmitting nodes choose to send simultaneously

Random access protocols

- A **random-access MAC protocol** specifies:
 - (1) How to detect collisions
 - (2) How to recover from collisions
 - ... Usually by **retransmitting** after a while
- Examples of random-access MAC protocols:
 - Slotted ALOHA
 - ALOHA
 - CSMA
 - CSMA/CD
 - CSMA/CA

Carrier Sensing Multiple Access (CSMA)

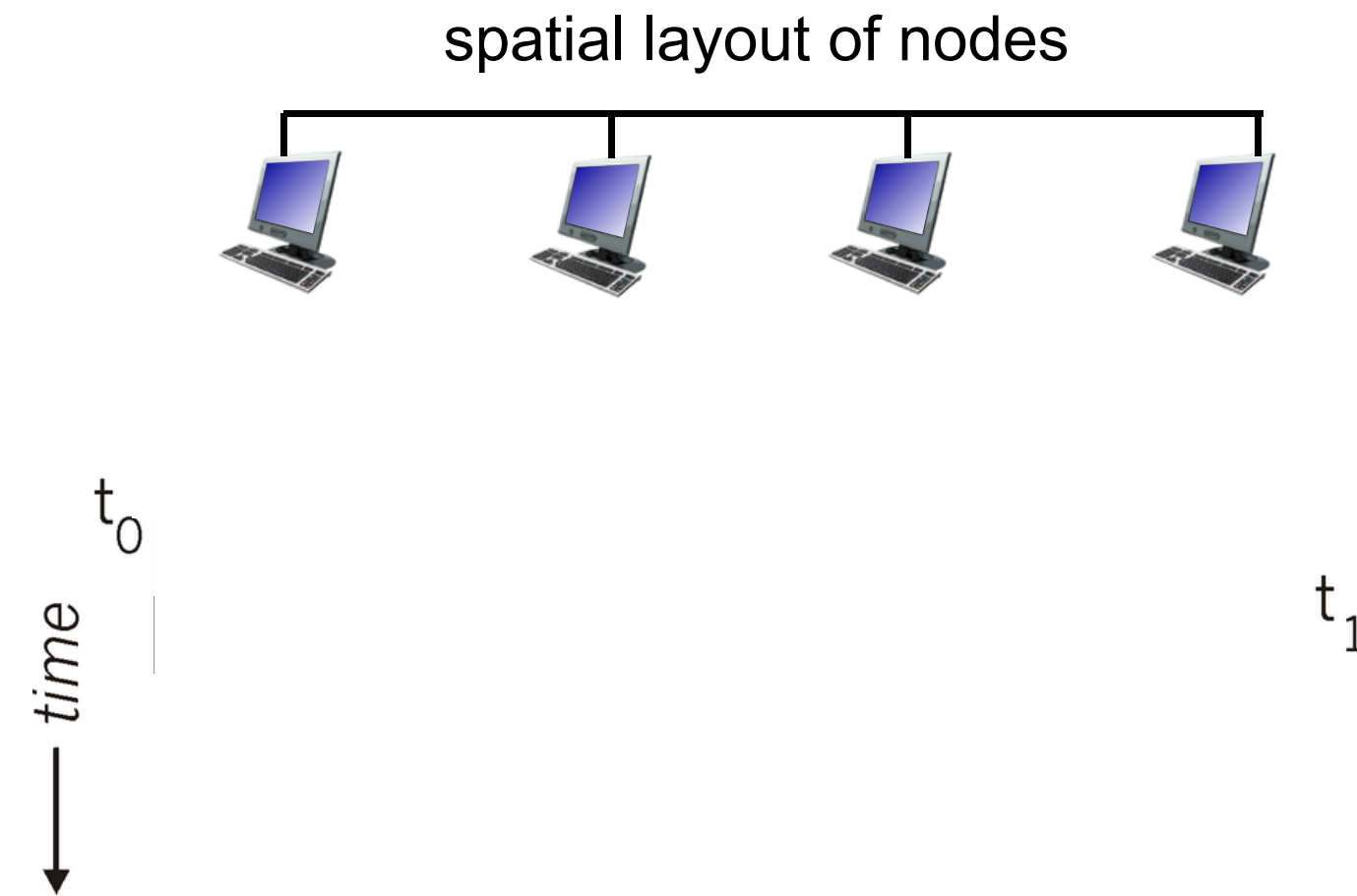
Carrier Sense Multiple Access (CSMA)

Key idea: **listen to the channel before transmission**

- If the channel sensed idle, then transmit entire frame
- If the channel sensed busy, then defer transmission
- Human analogy: don't interrupt others!

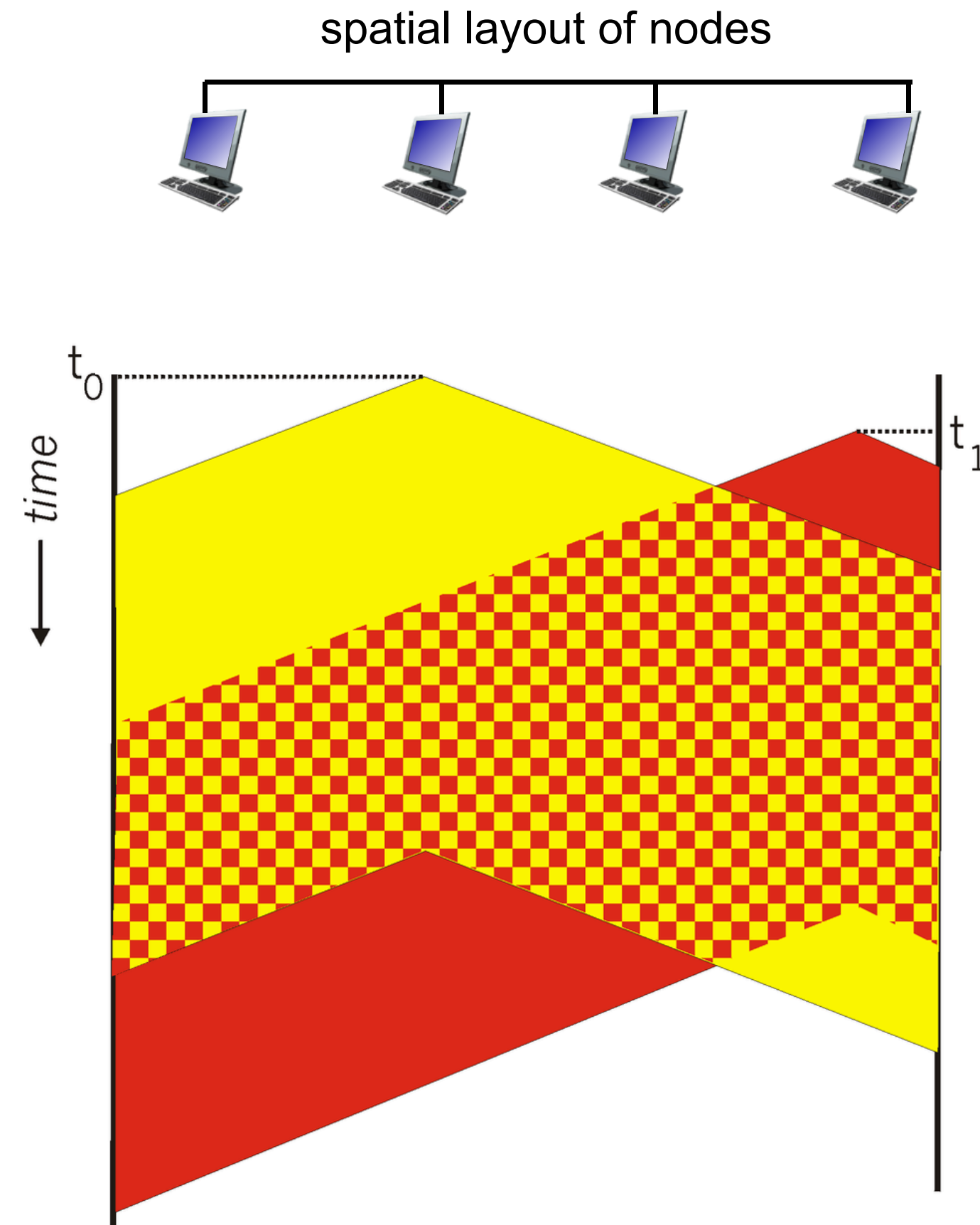
CSMA collisions

- Collisions can still occur.
Due to propagation delay, two nodes may not hear each other's transmissions right away
- If there is a collision, the entire packet transmission time is wasted



CSMA collisions

- **Collisions can still occur.**
Due to propagation delay, two nodes may not hear each other's transmissions right away
- If there is a collision, the entire packet transmission time is wasted
- **Propagation delay** between the nodes plays a key role in determining the collision probability



Carrier Sensing with Collision Detection (CSMA/CD)

Carrier sensing + collision detection

- On certain types of link technologies (e.g., wired Ethernet), continuous carrier sensing can enable **detecting collisions** very quickly
 - Not always possible, e.g., wireless LANs (next lecture)
- Suppose it is possible to detect collisions **as soon as signal from one node reaches another (transmitting) node**
 - Measure received signal, compare with transmitted signal
- CSMA/CD: **Stop transmitting when a collision is detected**
 - Reduce wastage of channel capacity
 - Improve the chances of retransmitting earlier
- Analogy: “polite speaker”: stop speaking when interrupted

CSMA/CD

- **Collisions can still occur.** Due to propagation delay, two nodes may not hear each other's transmissions right away
- However, upon detecting a collision, node stops transmitting

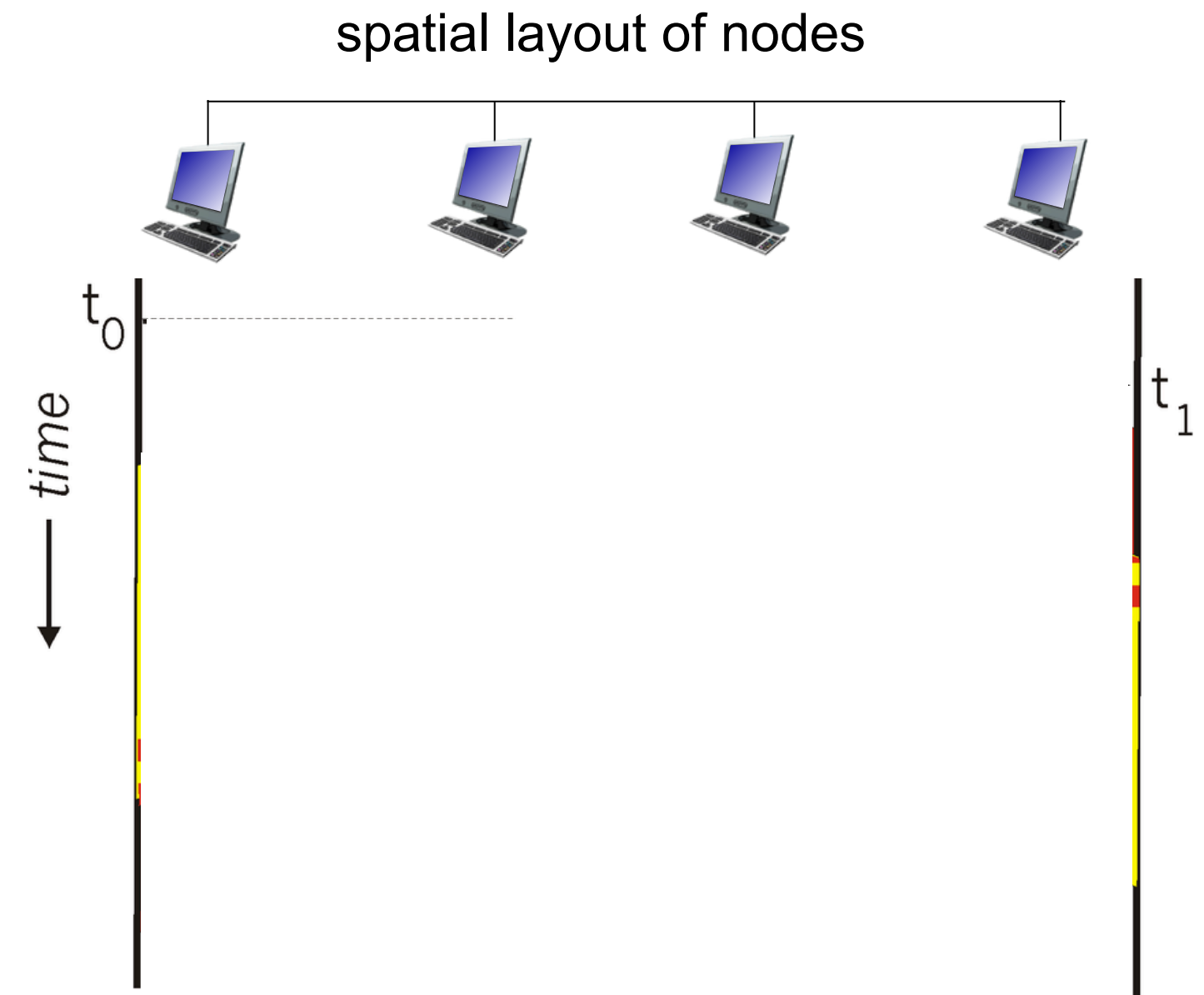


Figure 12.8 *Spacetime model of the collision in CSMA*

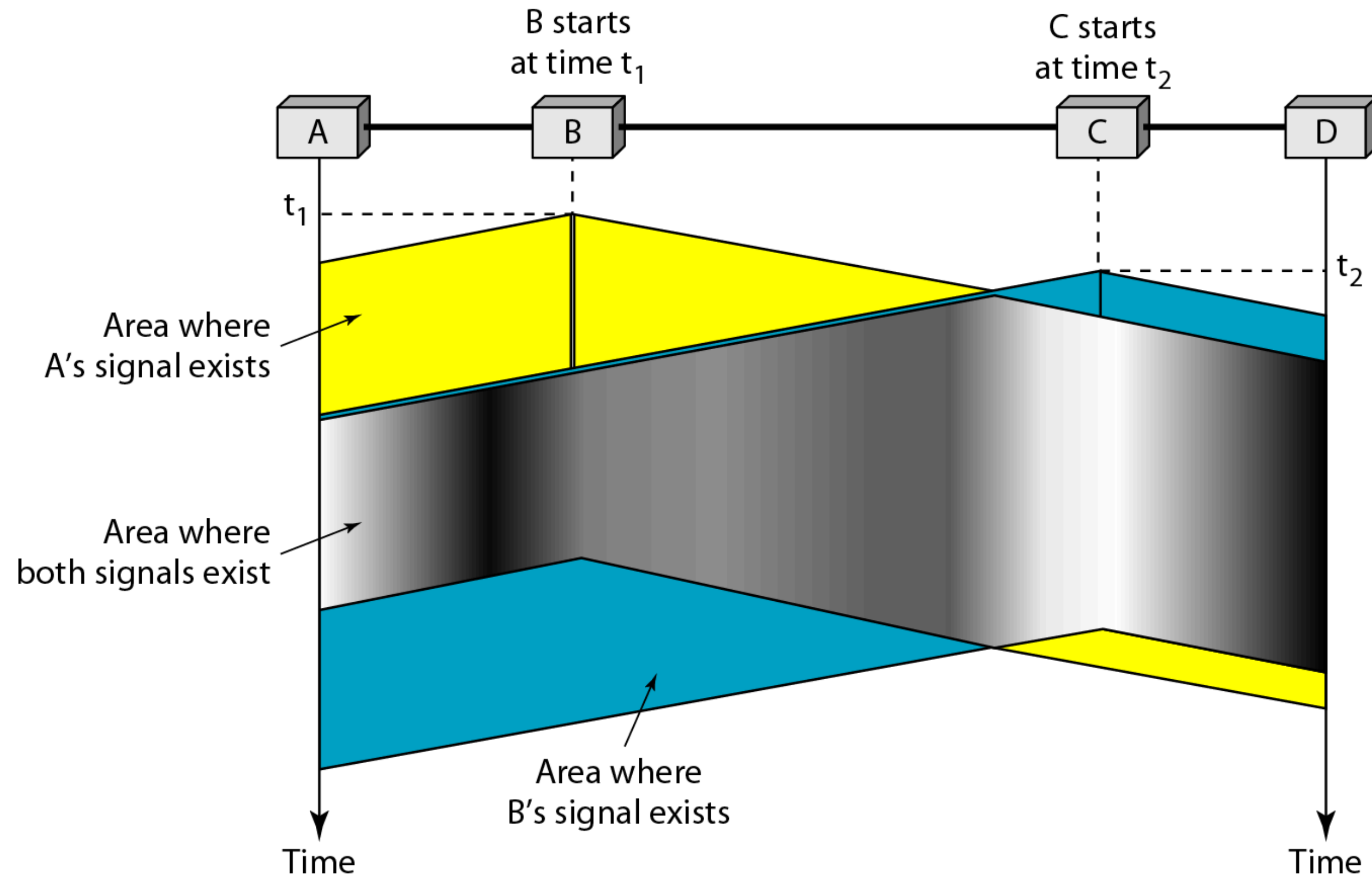


Figure 12.9 *Vulnerable time in CSMA*

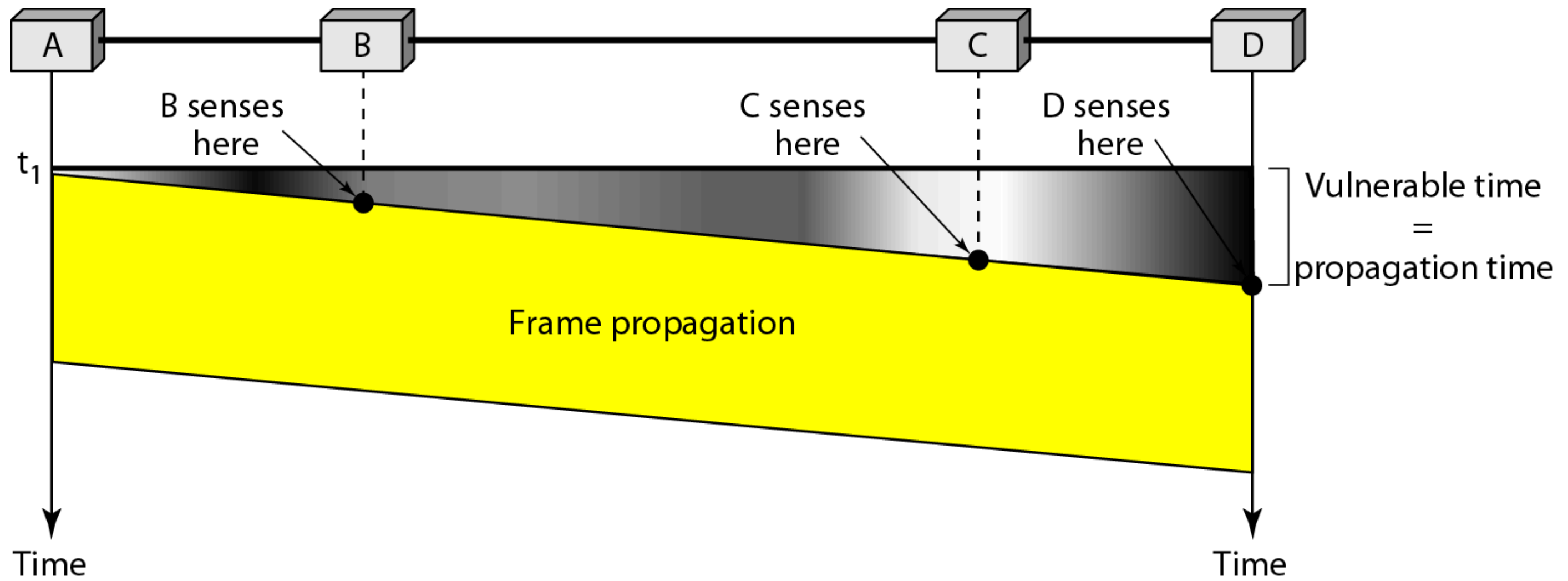
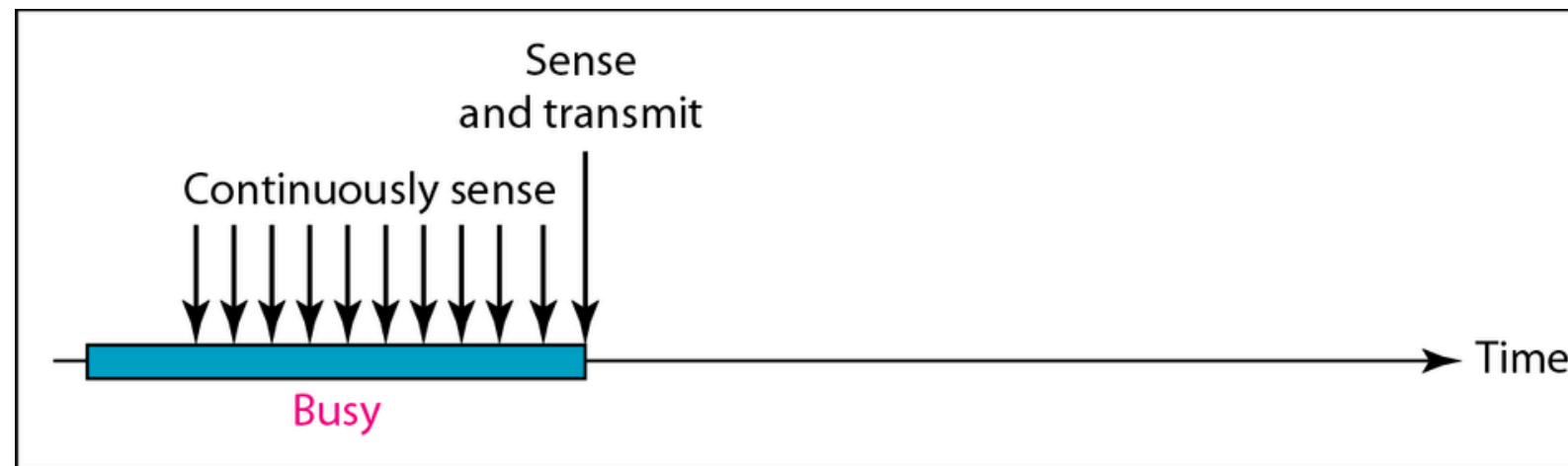
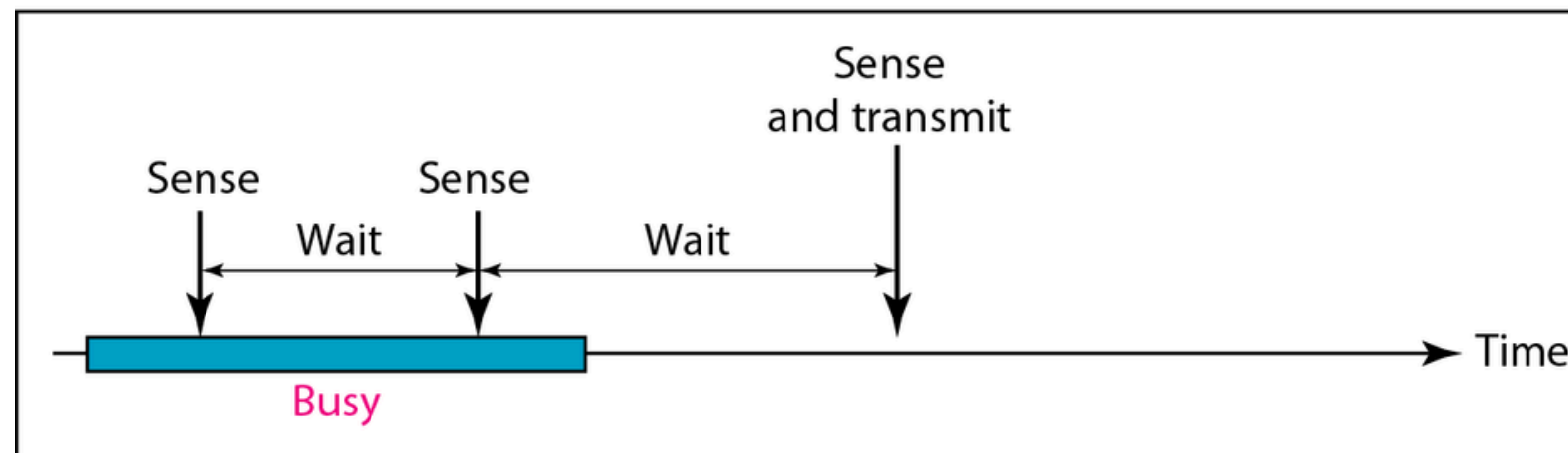


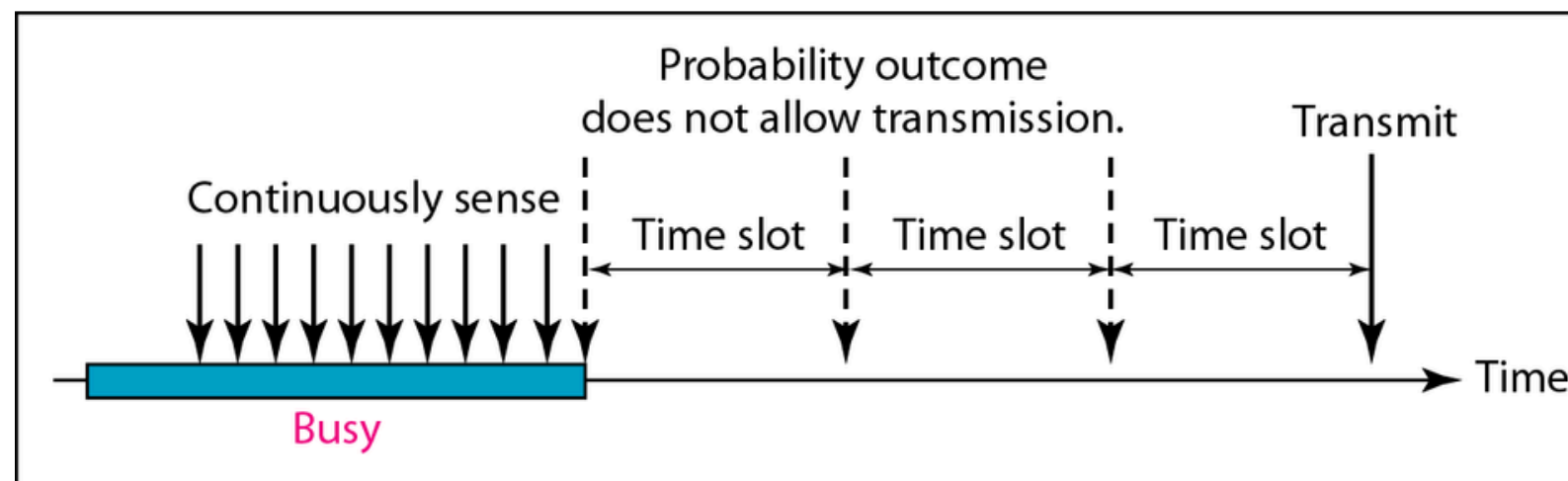
Figure 12.10 *Behavior of three persistence methods*



a. 1-persistent



b. Nonpersistent



c. p-persistent

Figure 12.11 *Flow diagram for three persistence methods*

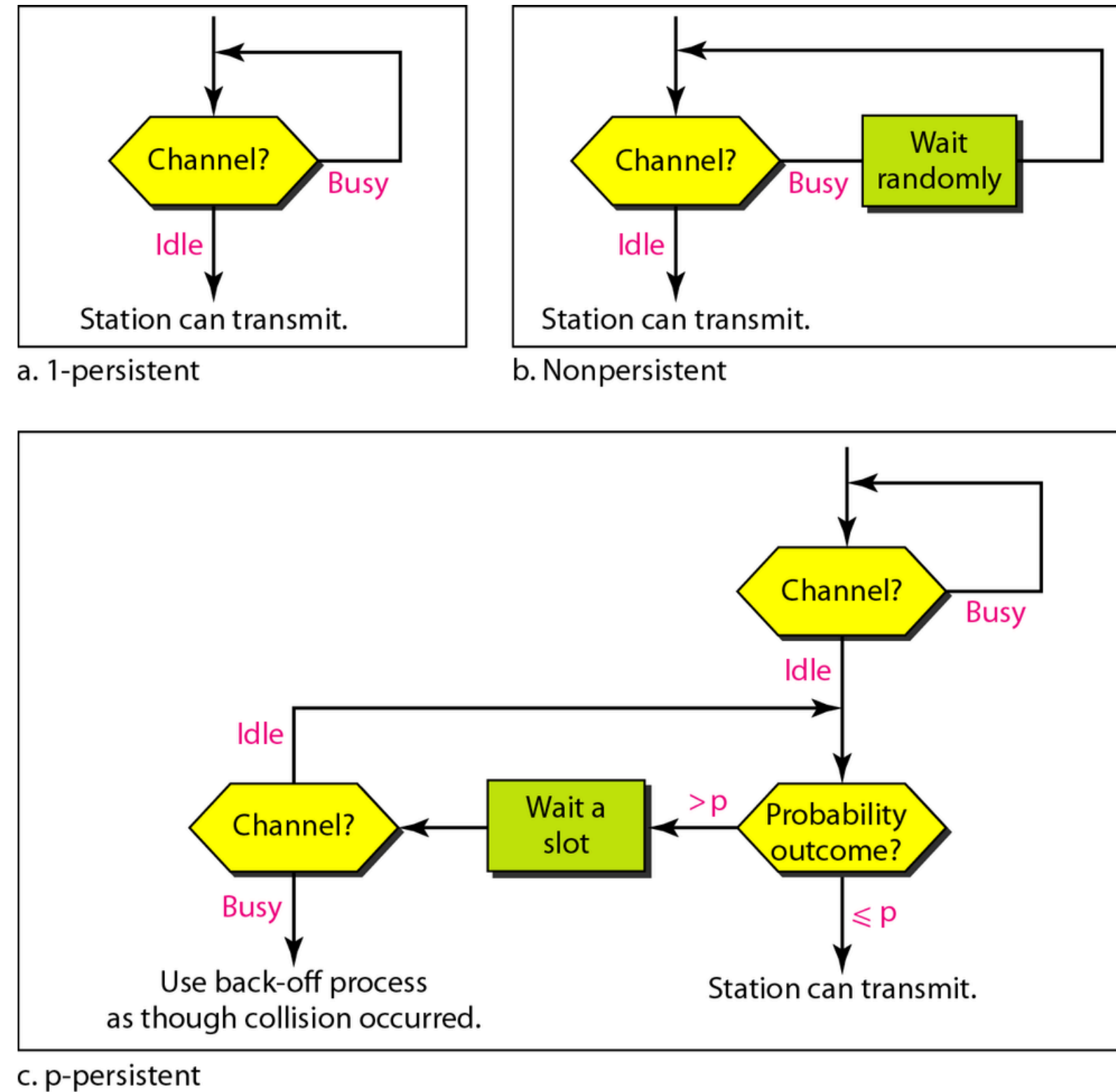


Figure 12.12 *Collision of the first bit in CSMA/CD*

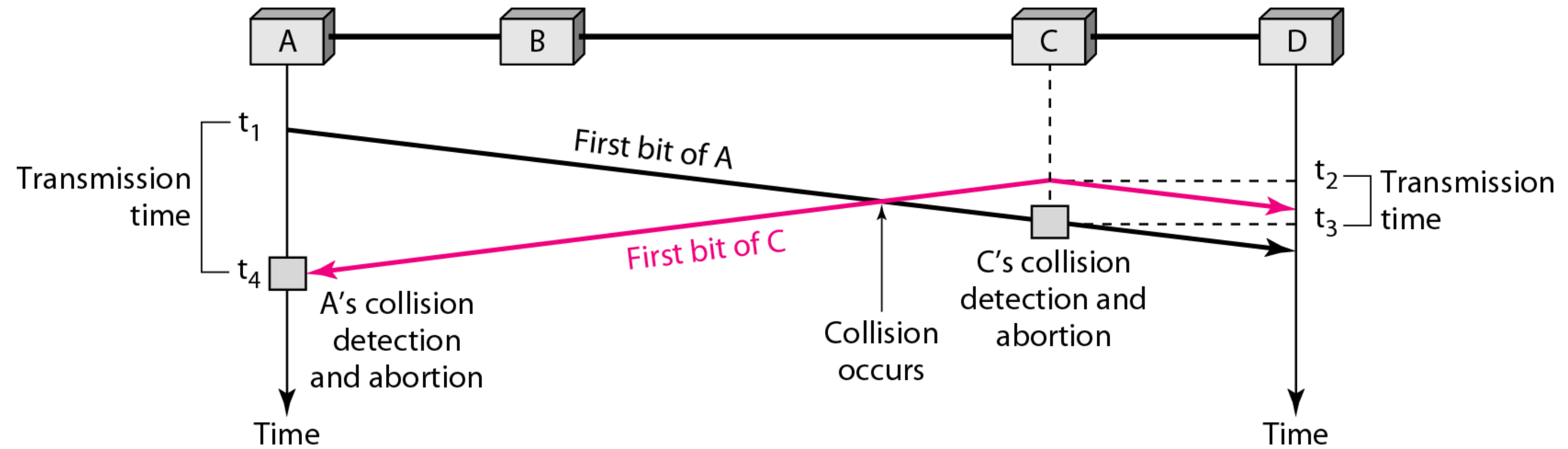


Figure 12.13 *Collision and abortion in CSMA/CD*

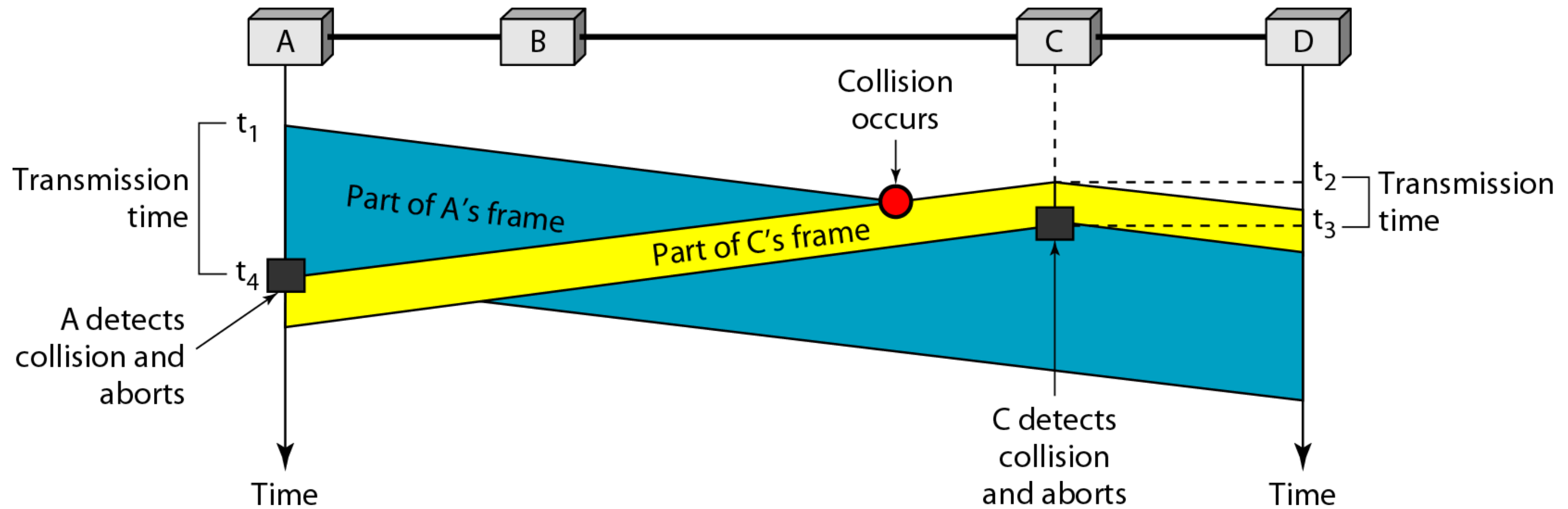


Figure 12.14 *Flow diagram for the CSMA/CD*

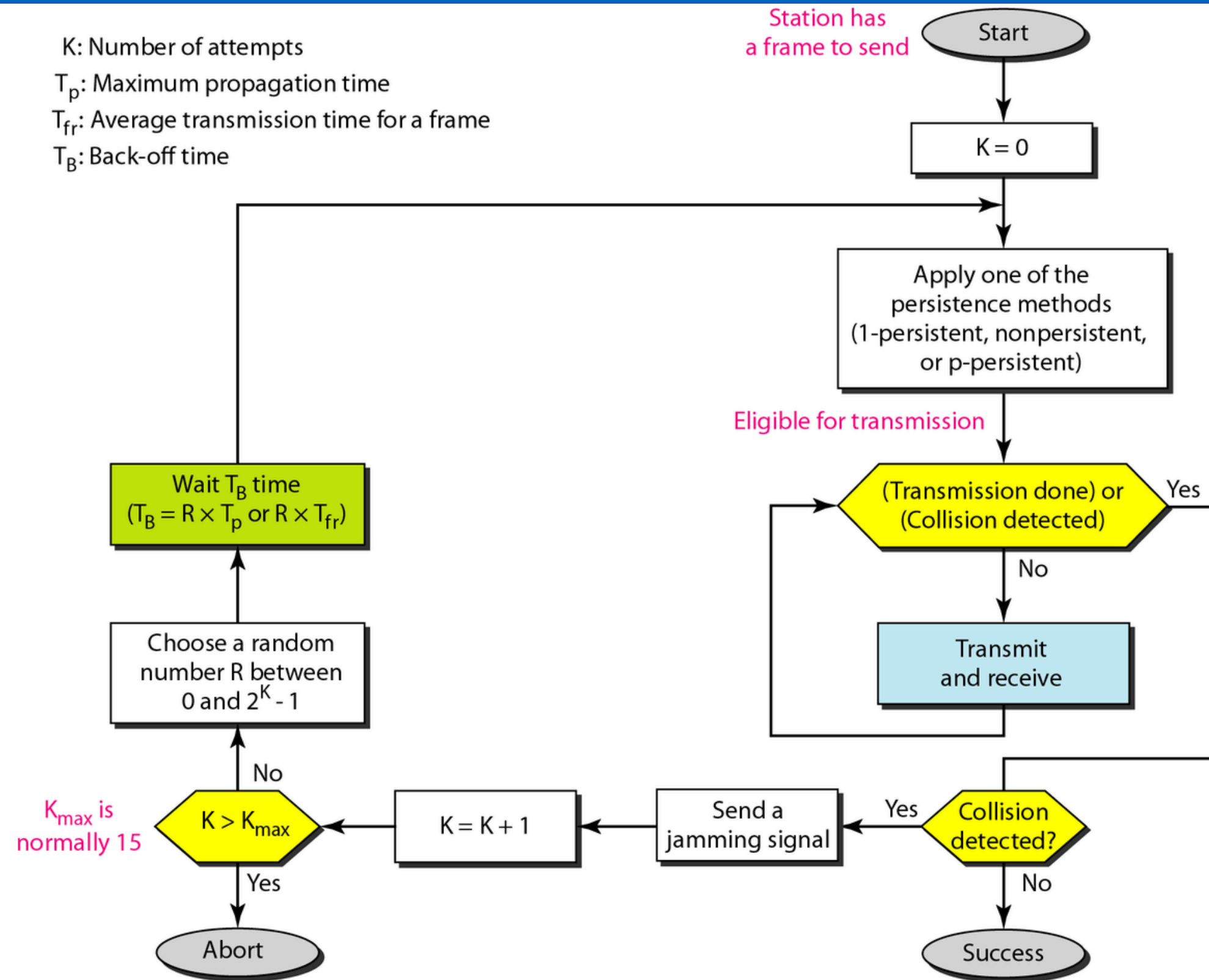
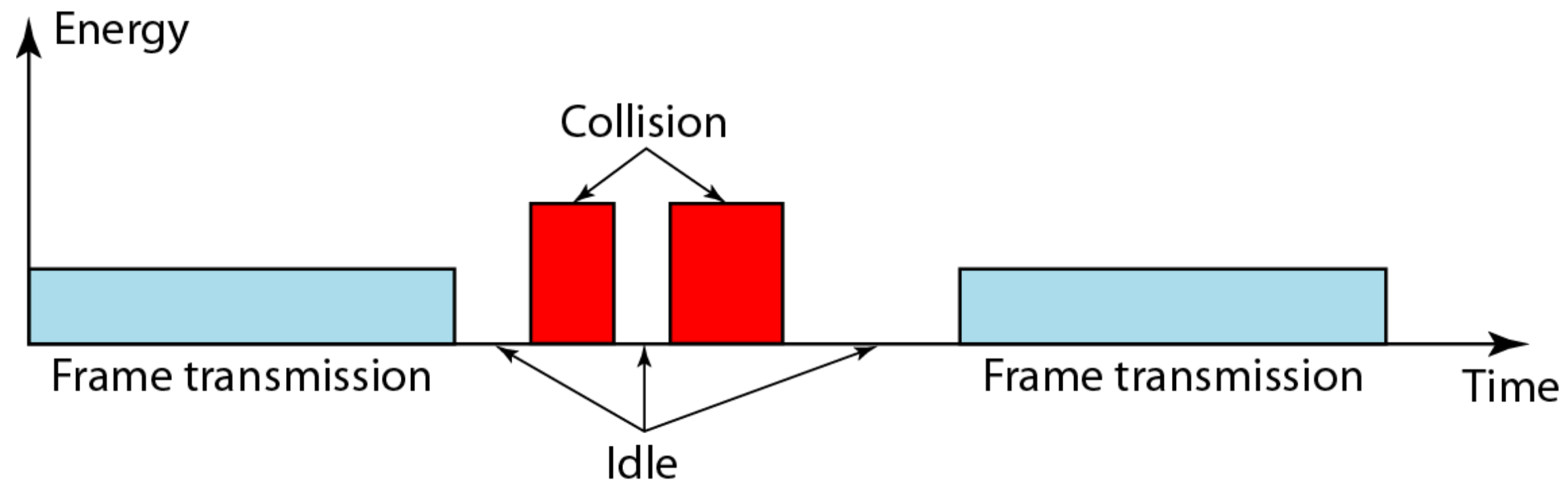


Figure 12.15 *Energy level during transmission, idleness, or collision*



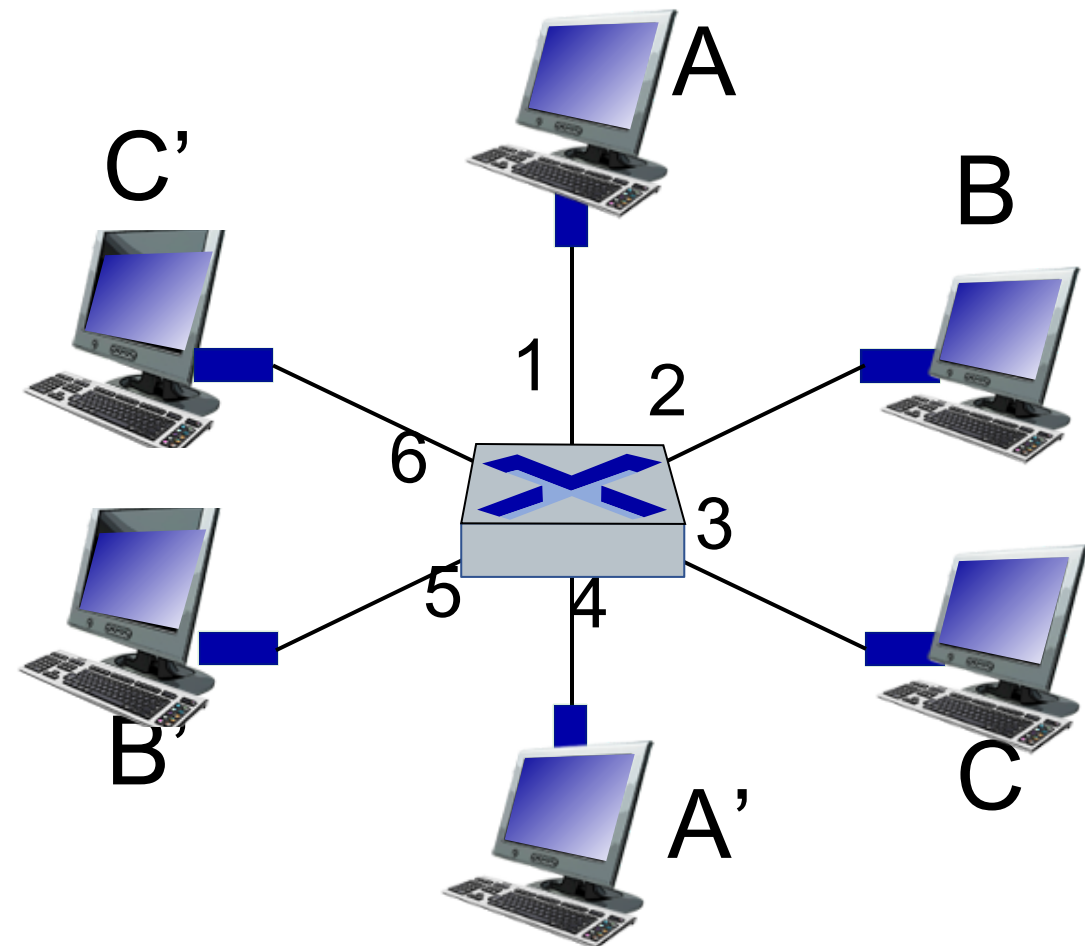
Review of CSMA/CD

- Sense the medium for existing transmissions
- If the medium is idle, transmit
- If a collision is detected during transmission, abort the transmission
- Retry transmission “later”
- Retransmission in the ALOHA protocols:
 - Transmit with some probability during the next frame time

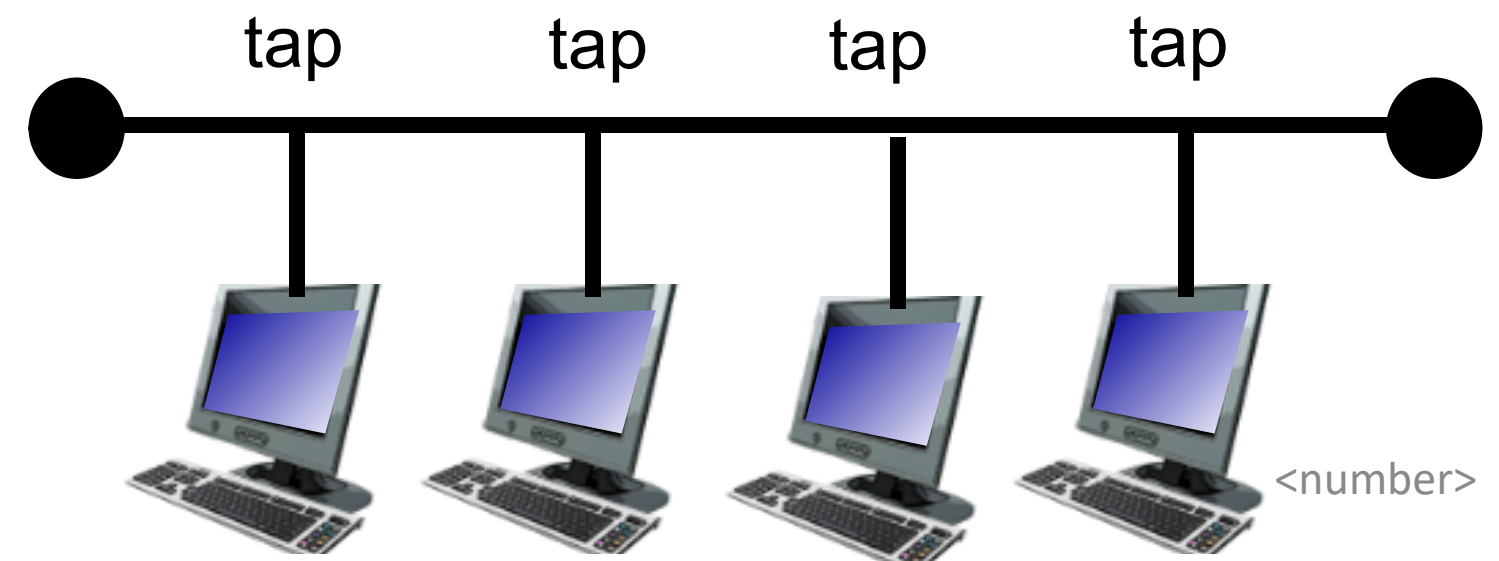
CSMA/CD in Shared Ethernet

Switched Ethernet vs. Shared Ethernet

- Nodes connected over **dedicated link** to a switch (previous lecture)



- Multiple nodes connected to the **same Ethernet cable**
- Classic method: limited geographic coverage, easy to add nodes by adding a **tap**
- Every node can hear every other node



MAC in shared Ethernet

- Since the cable is a broadcast medium, it is important to decide which node can transmit at any time.
- Shared Ethernet uses CSMA/CD MAC, with a twist

Shared Ethernet: CSMA/CD

1. NIC receives data to send (e.g., from network layer), creates a frame
2. If NIC **senses channel idle**, starts frame transmission. If NIC senses channel busy, waits until channel idle, then transmits (**CSMA**)
3. If NIC transmits entire frame without detecting another transmission, NIC is done with frame.
4. If NIC detects another transmission while transmitting, the NIC aborts the transmission and sends a “jam signal” (**CD**)
5. The NIC waits **for some period** and attempts to retransmit the frame.
(Note: in ALOHA, retransmission is probabilistic)
Q: how long should the NIC wait?

Reference

- **Chapter: 12, Data Communications and Networking**, Fourth Edition by Behrouz A. Forouzan